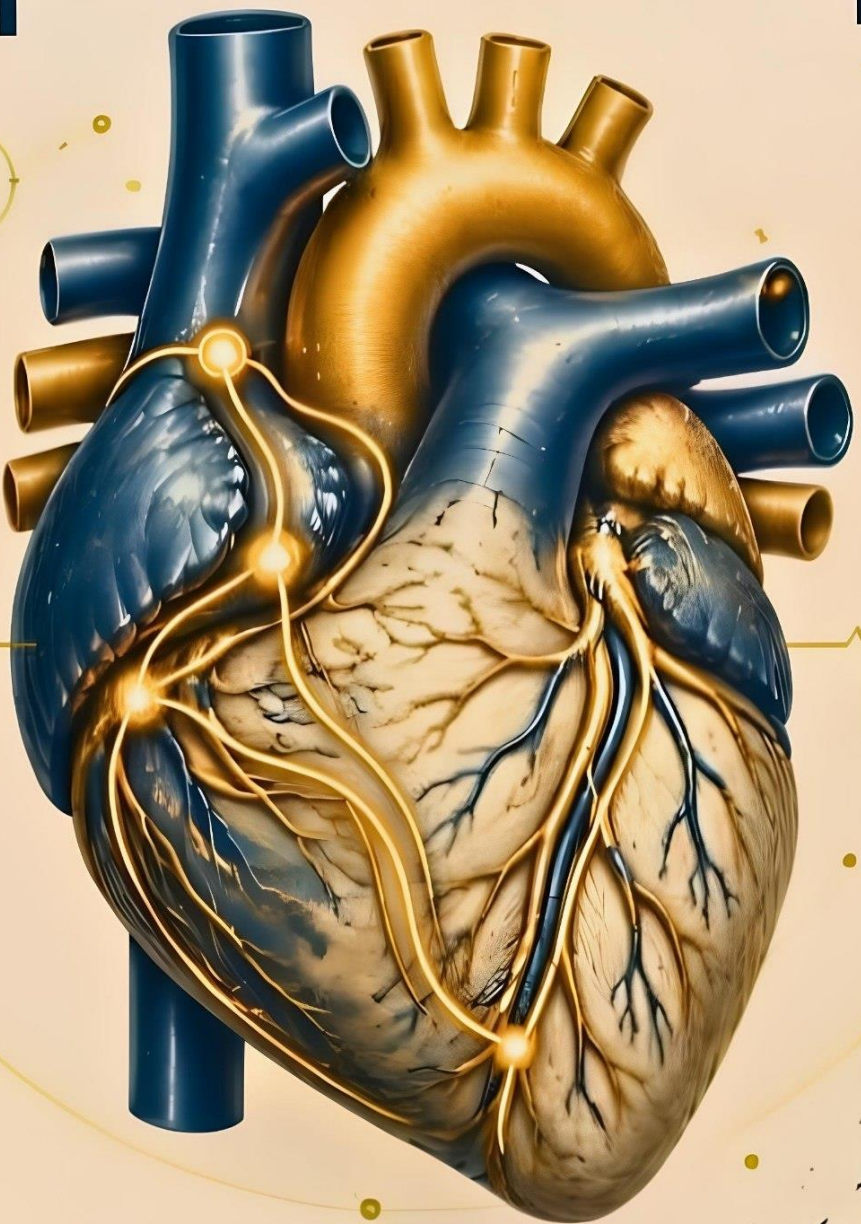
PHYSIOLOGY

Part 1

MCQS



2026

بسم الله الرحمن الرحيم

والصلاة والسلام على خير المرسلين محمد بن عبد الله وعلى آله وصحبه وسلم أما بعد، نقدم بين يديك ملف

(General Physiology MCQ)

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ابدأ مستعيناً بالله وكرر الحل ما استطعت واحرص على وقتك فهو رأس مالك وأغلى ما تمل

حسابات الفريق على منصات التواصل الاجتماعي

جميع الحقوق محفوظة لفريق مِعوان

ولا نحلل بيعها بعائد ربحي أو سرقتها بأي شكل من الأشكال

Body Fluids

Pre Exams & Formative

1-	The most abundant cation in ICF is (MID 31, MID 34, Formative 36)?						
A-	Sodium	B-	Magnesium	C-	Potassium	D-	Calcium

2-	The major extracellular anion is (FINAL 32)?						
A-	potassium	B-	Sodium	C-	chloride	D-	magnesium

3-	Extracellular fluid in adults differs from intracellular fluid in that its (2ND TRIAL 32)?		
A-	Volume is greater	B-	Tonicity is lower
C-	Anions are mainly inorganic	D-	Sodium: potassium molar ratio is higher

4-	Compared to the intracellular fluid, the extracellular fluid has a high concentration of (MID 33)?						
A-	Protein	B-	Phosphate	C-	Sodium	D-	Potassium

5-	Concerning venous end capillaries (FINAL 34)?
A-	Hydrostatic pressure at capillaries equal to osmotic pressure af protien in plasma
B-	hydrostatic pressure st venous end equal to hydrostatic pressure at arterial end
C-	hydrostatic pressure at venous end is less than osmotic pressure af plasma protein

6-	The term oncotic pressure is applied to osmotic pressure exerted (FINAL 34)?
A-	albumin and other plasma proteins across the capillary wall
B-	Crystalloids across the cell membrane
C-	Hemoglobin across the capillary wall
D-	Substances such as urea , lactate , glucose , across the cell membrane

7-	The total water content in the body represents ..... of the total body weight of the normal adult male (2ND TRIAL 35)?						
A-	80%	B-	60%	C-	40%	D-	20%

8-	The intracellular fluid has a higher concentration of ..... in comparison with extracellular fluid (2ND TRIAL 35)?						
A-	Sodium	B-	Potassium	C-	Chloride	D-	Calcium

9-	Regarding Osmolarity of different body fluids: (MID 37)?		
A-	Osmotic pressure is 290-300 osmol/L	B-	Osmotic pressure is 290-300 osmol/dL
C-	Osmotic pressure is 290-300 osmol/ml	D-	Osmotic pressure is 290-300 mosmol/L

10-	Which is responsible for Keep Osmotic pressure (FINAL 37)?		
A-	Plasma protein in blood capillary	B-	Na ions concentration in ECF

11-	Regarding body fluids, which of the following is correct:- (Formative 36)?
A-	Oedema can occur if the plasma colloidal osmotic pressure increases
B-	The total extracellular fluid can be measured using the dilution method of Evans blue
C-	The percentage of the total body water decreases with age
D-	The volume of interstitial fluid is more than the volume of intracellular fluid

12-	the main absorptive force at the venular end during interstitial fluid formation is (Formative 36)?		
A-	plasma protein osmotic pressure	B-	capillary hydrostatic pressure
C-	interstitial fluid osmotic pressure	D-	interstitial fluid pressure

13-	the total water content in the body represents of the total body weight of the normal adult (2ND TRIAL 37)?						
A-	80%	B-	60%	C-	40%	D-	20%

14-	Which of the following has a higher concentration in plasma than in interstitial fluid? (2ND TRIAL 37)
A-	Protein
B-	Glucose
C-	Potassium
D-	Sodium

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
C	C	D	C	C	A	B	B	D	B
11	12	13	14	15	16	17	18	19	20
C	A	B	A	-	-	-	-	-	-

Department Qs

15-	the total water content in the body represents of the total body weight of the normal adult (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	40%	D-	20%

16-	the total water content in the body equals ..... in a person 70 kg (SNU Bank)?						
A-	40 L	B-	28 L	C-	14 L	D-	10 L

17-	the total water content in the body equals ..... in a person 70 kg (Dr. Mohamed Ashraf)?						
A-	42 L	B-	28 L	C-	14 L	D-	10 L

18-	the total water content in the intracellular fluid represents .... of the total body weight of the normal adult (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	40%	D-	20%

19-	the total water content in the intracellular fluid equals ..... in a person 70 kg (SNU Bank, Dr.Mohamed Ashraf)?						
A-	40 L	B-	28 L	C-	14 L	D-	10 L

20-	the total water content in the extracellular fluid represents of the total body weight of the normal adult (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	40%	D-	20%

21-	the total water content in the extracellular fluid equals in a person 70 kg (SNU Bank, Dr.Mohamed Ashraf)?						
A-	42 L	B-	28 L	C-	14 L	D-	10 L

22-	the total water content in the interstitial spaces equals in a person 70 kg (SNU Bank, Dr.Mohamed Ashraf)?						
A-	42 L	B-	28 L	C-	14 L	D-	10 L

23-	the total water content in the plasma equals in a person 70 kg (SNU Bank, Dr.Mohamed Ashraf)?								
A-	42 L	B-	28 L	C-	14 L	D-	10 L	E-	3 L

24-	the total water content in the transcellular fluid in a person 70 kg (SNU Bank, Dr.Mohamed Ashraf)?						
A-	14 L	B-	10 L	C-	3 L	D-	0.5 L

25-	the total water content in the body represents of the total body weight of the infants (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	40%	D-	20%

26-	the total water content in the body represents of the total body weight of the old age (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	45%	D-	20%

27-	the total water content in the body represents of the total body weight of the normal adult female (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	50%	D-	20%

28-	the total water content in the body represents of the total body weight of the obese persons (SNU Bank, Dr.Mohamed Ashraf)?						
A-	80%	B-	60%	C-	50%	D-	20%

29-	which of the following is in higher concentration in plasma than interstitial space: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	sodium	B-	K	C-	glucose	D-	proteins

30-	ICF and ECF are separated by: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	cell membrane	B-	capillary wall
C-	tight junctions	D-	not separated from each other

31-	plasma and interstitial fluid are separated by: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	cell membrane	B-	capillary wall
C-	tight junctions	D-	not separated from each other

32-	the concentration of protein in plasma is ..... their concentration in interstitial space: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	higher than	B-	lower than	C-	equal to	D-	variable

33-	the concentration of glucose in plasma is ..... their concentration in interstitial space: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	higher than	B-	lower than	C-	equal to	D-	variable

34-	The intracellular fluid has a higher concentration of ..... In comparison with extracellular fluid (SNU Bank, Dr.Mohamed Ashraf)?						
A-	sodium	B-	potassium	C-	chloride	D-	proteins

35-	The extracellular fluid has a higher concentration of ..... In comparison with intracellular fluid (SNU Bank, Dr.Mohamed Ashraf)?						
A-	sodium	B-	potassium	C-	phosphate	D-	sulphate

36-	calcium ions concentration is lowest in: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	intracellular fluid	B-	interstitial fluid	C-	plasma	D-	transcellular fluid

37-	the main filtration force of fluids from the plasma to the interstitial fluid is: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	capillary hydrostatic pressure	B-	plasma colloidal osmotic pressure
C-	interstitial fluid hydrostatic pressure	D-	interstitial fluid colloidal osmotic pressure

38-	which of the following causes non-pitting oedema: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	heart failure	B-	venous obstruction by thrombosis
C-	liver failure	D-	lymphatic obstruction

39-	which of the following causes oedema through increase in capillary hydrostatic pressure: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	allergy or inflammation	B-	venous obstruction by thrombosis
C-	liver failure	D-	lymphatic obstruction

40-	which of the following causes oedema through decrease in plasma colloidal osmotic pressure: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	allergy or inflammation	B-	venous obstruction by thrombosis
C-	liver failure	D-	lymphatic obstruction

41-	the capillary wall is NOT permeable to (Dr. Mohamed Ashraf)?						
A-	sodium	B-	proteins	C-	glucose	D-	water

42-	the main reabsorbing force of fluids from interstitial fluid to the plasma is: (Dr. Mohamed Ashraf)?		
A-	capillary hydrostatic pressure	B-	plasma colloidal osmotic pressure
C-	interstitial fluid hydrostatic pressure	D-	interstitial fluid colloidal osmotic pressure

43-	which of the following causes oedema through increase in capillary permeability: (Dr. Mohamed Ashraf)?		
A-	allergy or inflammation	B-	venous obstruction by thrombosis
C-	liver failure	D-	lymphatic obstruction

44-	Which of the following causes generalized oedema: (Dr. Mohamed Ashraf)?		
A-	trauma to the hand	B-	venous obstruction
C-	pregnancy	D-	liver disease

45-	which of the following causes localized oedema: (Dr. Mohamed Ashraf)?						
A-	pregnancy	B-	heart failure	C-	liver failure	D-	renal failure

46-	The mass movement of fluids into and out of capillary beds depending on the pressure gradient is: (Dr. Mohamed Ashraf)?						
A-	Diffusion	B-	Bulk flow	C-	Filtration	D-	Reabsorption

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	A	A	C	B	D	C	D	E	D
11	12	13	14	15	16	17	18	19	20
A	C	C	C	D	A	B	A	C	B
21	22	23	24	25	26	27	28	29	30
A	A	A	D	B	C	B	B	A	D
31	32	33	34	35	36	37	38	39	40
A	B	-	-	-	-	-	-	-	-

M3WAN Bank

1-	Total body water in human beings normally varies between: (M3WAN - Core)?		
A-	45% and 75% of body weight	B-	10% and 30% of body weight
C-	80% and 90% of body weight	D-	20% and 40% of body weight

2-	Total body water in an average 70 kg adult approximates: (Important M3WAN - Core)?						
A-	14 L	B-	28 L	C-	40 L	D-	50 L

3-	Body fluids are important in all of the following EXCEPT: (M3WAN - Core)?		
A-	Homeostasis	B-	Temperature regulation
C-	Metabolic reactions	D-	Storage of hereditary genetic information

4-	Regarding the effect of age on total body water content: (M3WAN - Core)?
A-	Water content decreases in old age due to increased adipose tissue
B-	Newly born infants have about 45% body water
C-	Old age is associated with about 80% body water
D-	Body water content does not change with age

5-	Compared with adult males, adult females have: (M3WAN - Core)?
A-	Lower total body water due to more subcutaneous fat
B-	Higher total body water due to less subcutaneous fat
C-	Equal total body water
D-	Higher total body water due to more muscle mass

6-	In obese individuals, total body water approximates what percentage of body weight? (M3WAN - Hard)?						
A-	30%	B-	65%	C-	50%	D-	80%

7-	Intracellular fluid represents what fraction of total body water? (M3WAN - Core)?						
A-	1/3	B-	2/3	C-	1/4	D-	1/2

8-	Intracellular fluid represents approximately what fraction of total body weight? (M3WAN - Core)?						
A-	5%	B-	20%	C-	15%	D-	40%

9-	Extracellular fluid represents what fraction of total body water? (M3WAN - Core)?						
A-	2/3	B-	1/3	C-	1/4	D-	3/4

10-	Extracellular fluid volume in a 70 kg adult approximates: (M3WAN - Core)?						
A-	3 L	B-	10.5 L	C-	14 L	D-	28 L

11-	Plasma represents what fraction of extracellular fluid? (M3WAN - Core)?						
A-	1/2	B-	1/4	C-	1/3	D-	3/4

12-	The largest component of extracellular fluid by volume is: (M3WAN - Core)?						
A-	Plasma	B-	Transcellular fluid	C-	Lymph	D-	Interstitial fluid

13-	Transcellular fluid includes all of the following EXCEPT: (M3WAN - Hard)?		
A-	Cerebrospinal fluid	B-	Synovial fluid
C-	Plasma	D-	Intraocular fluid

14-	The capillary wall keeps the protein content of intravascular fluid higher than that of interstitial fluid mainly because: (M3WAN - Hard)?
A-	Interstitial fluid actively pumps proteins back to plasma
B-	The capillary wall is impermeable to all solutes
C-	Proteins are electrically repelled by the capillary wall
D-	Proteins are too large to pass through the capillary pores

15-	The predominant cation in extracellular fluid is: (M3WAN - Core)?						
A-	Magnesium	B-	Sodium	C-	Potassium	D-	Phosphate

16-	The predominant cation in intracellular fluid is: (Important M3WAN - Core)?						
A-	Sodium	B-	Calcium	C-	Potassium	D-	Magnesium

17-	**The major anions of intracellular fluid include: (M3WAN - Hard)?**		
A-	Chloride and bicarbonate	B-	Sodium and calcium
C-	Bicarbonate and sodium	D-	Phosphates and proteins

18-	**According to the indicator dilution method (Fick's principle), the amount of an injected indicator substance equals: (M3WAN - Core)?**
A-	Concentration only, regardless of volume
B-	Volume of the compartment divided by its concentration
C-	Volume of the compartment multiplied by its concentration
D-	The rate of substance excretion

19-	**An ideal substance used for measuring a body fluid compartment by the indicator dilution method should be: (M3WAN - Core)?**
A-	Rapidly excreted
B-	Toxic at high concentration
C-	Utilized by tissues
D-	Uniformly distributed in the compartment measured

20-	**Which of the following body fluid compartments is calculated by subtraction rather than measured directly by the dilution method? (M3WAN - Hard)?**		
A-	Plasma volume	B-	Extracellular fluid volume
C-	Total body water	D-	Intracellular fluid volume

21-	**Plasma volume is best measured using: (M3WAN - Core)?**		
A-	Inulin	B-	Deuterium oxide
C-	Evans blue dye	D-	Radioactive sodium

22-	**Red blood cell volume can be measured using red cells labeled with: (M3WAN - Core)?**		
A-	Evans blue dye	B-	Inulin
C-	Deuterium oxide	D-	Radioactive chromium (51Cr)

23-	**Blood volume can be calculated from plasma volume and the hematocrit value (HV) using the formula: (M3WAN - Hard)?**
A-	Blood volume = plasma volume × HV / 100
B-	Blood volume = plasma volume × 100 / (100 – HV)
C-	Blood volume = plasma volume – HV
D-	Blood volume = plasma volume / HV

24-	Which substance is used to measure extracellular fluid volume? (M3WAN - Core)?		
A-	Evans blue dye	B-	Radioactive chromium
C-	Deuterium oxide	D-	Inulin

25-	Total body water is measured using a substance that: (M3WAN - Core)?
A-	Cannot cross the capillary wall
B-	Is confined to the vascular compartment
C-	Crosses both the capillary wall and the cell membrane
D-	Is confined to the extracellular fluid

26-	The body fluid compartment volume that holds an injected indicator substance is called the: (M3WAN - Core)?		
A-	Osmotic volume	B-	Distribution volume
C-	Reference volume	D-	Clearance volume

27-	The main function of tissue (interstitial) fluid is to: (M3WAN - Core)?
A-	Store hereditary information
B-	Generate action potentials
C-	Serve as the medium for exchange of substances between cells and capillaries
D-	Synthesize plasma proteins

28-	Intracellular fluid volume is calculated as: (M3WAN - Hard)?
A-	Extracellular fluid volume minus plasma volume
B-	Total body water plus extracellular fluid volume
C-	Total body water minus extracellular fluid volume
D-	Plasma volume minus interstitial fluid volume

29-	Capillary hydrostatic pressure at the venous end of the capillary approximates: (Important M3WAN - Core)?						
A-	35 mmHg	B-	10 mmHg	C-	25 mmHg	D-	17 mmHg

30-	Plasma colloidal osmotic pressure at the arterial and venous ends of the capillary is: (Important M3WAN - Core)?		
A-	Higher at the arterial end	B-	Higher at the venous end
C-	Equal at both ends, about 25 mmHg	D-	Zero at both ends

31-	**Net filtration occurs at the arterial end of the capillary because: (M3WAN - Hard)?**
A-	Oncotic pressure exceeds hydrostatic pressure
B-	Hydrostatic pressure exceeds oncotic pressure
C-	Hydrostatic and oncotic pressures are equal
D-	Oncotic pressure is zero

32-	**Oedema is best defined as: (M3WAN - Core)?**
A-	Deficiency of fluid in the vascular compartment
B-	Excess protein in the plasma
C-	Presence of excess fluid in the interstitial space
D-	Presence of excess fluid in the intracellular space

33-	**Oedema due to right-sided heart failure results mainly from: (M3WAN - Core)?**		
A-	Decreased plasma oncotic pressure	B-	Increased capillary hydrostatic pressure
C-	Lymphatic obstruction	D-	Increased capillary permeability

34-	**Increased capillary hydrostatic pressure leading to oedema may result from: (M3WAN - Core)?**		
A-	Bacterial toxins	B-	Decreased synthesis of plasma proteins
C-	Venous obstruction by thrombosis	D-	Lymphatic obstruction

35-	**Oedema of the lower limbs during pregnancy may result mainly from: (M3WAN - Hard)?**
A-	Increased capillary permeability
B-	Compression on the veins from outside by the enlarging uterus
C-	Decreased plasma oncotic pressure
D-	Lymphatic obstruction by filariasis

36-	**Hypoproteinemic oedema may result from: (M3WAN - Core)?**		
A-	Venous thrombosis	B-	Filariasis
C-	Bacterial toxins	D-	Severe kidney disease with protein loss

37-	**Decreased synthesis of plasma proteins leading to hypoproteinemic oedema may occur in: (M3WAN - Core)?**		
A-	Liver cell failure	B-	Severe kidney disease
C-	Filariasis	D-	Bacterial infection

38-	**Non-pitting oedema with high interstitial protein content is characteristic of: (Important M3WAN - Hard)?**		
A-	Hypoproteinemia	B-	Right-sided heart failure
C-	Increased capillary permeability	D-	Lymphatic obstruction

39-	Increased capillary permeability leading to oedema is typically caused by: (M3WAN - Core)?		
A-	Venous obstruction	B-	Bacterial toxins or allergic conditions
C-	Undernutrition	D-	Filariasis

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
A	C	D	A	A	C	B	D	B	C
11	12	13	14	15	16	17	18	19	20
B	D	C	D	B	C	D	C	D	D
21	22	23	24	25	26	27	28	29	30
C	D	B	D	C	B	C	C	D	C
31	32	33	34	35	36	37	38	39	40
B	C	B	C	B	D	A	D	B	-

لَوْ عَلِمْتَ أَنَّ لَذَّةَ قَهرِ الهَوى أَطْيَبُ مِنْ مِثْلِهِ لَما غَلَبَكَ .

-ابن الجوزي | رحمه الله .

وفي قوة قهر الهوى لذةٌ تزيد على كل لذة .

ألا ترى إلى كل مغلوبٍ بالهوى كيف يكون ذليلاً ؟ لأنه قُهِرَ . . بخلاف غالِبِ الهوى، فإنه يكون

قويَّ القلبِ، عزيزاً .

-ابن الجوزي | رحمه الله .

غفلة ساعة تُذهِبُ مُجاهَدة سنة .

-الحسن البصري | رحمه الله .

"مَن كان لِعِنان هواه أملك، كان لطُرُق الرَّشاد أسلك ."

"ومَن اجتهد في مجاهدة هواه، فسيُكفَى أعدى عداه ." .

Homeostasis

Pre Exams & Formative

1-	The maintenance of an organism's internal environment is called (MID 31)?						
A-	Homeostasis	B-	Osmosis	C-	Cell transport	D-	Endocytosis

2-	Which of the following is NOT an example of the body homeostasis (MID 31)?
A-	Maintaining the body temperature of 98.6
B-	Body regulating sugar in the body
C-	Thinking
D-	The body making you thirsty when you are dehydrated

3-	PH balance is maintained by (MID 31)?						
A-	Kidney	B-	Liver	C-	Hypothalamus	D-	Brian

4-	Homeostasis in the human body is often maintained by a (FINAL 31)?		
A-	Negative feedback loop	B-	Brain loop
C-	Positive feedback loop	D-	Osmosis

5-	The "cause" for something to happen is the (FINAL 31)?						
A-	Stimulus	B-	Response	C-	Effect	D-	Center

6-	What component of all homeostatic control systems relays the information back to the control center (FINAL 31)?		
A-	Afferent	B-	Receptor
C-	Effector organ	D-	Coordinating center

7-	A regulatory mechanism in which the response to a stimulus, in a control system, causes the controlled variable to be in a steady state (MID 32)?		
A-	Extrinsic pathway	B-	Intrinsic pathway
C-	Feedback	D-	Inhibitory pathway

8-	A regulatory mechanism in which the response to a stimulus, in a control system, causes the controlled variable to move further a steady state (2ND TRIAL 32)?		
A-	Extrinsic pathway	B-	Intrinsic pathway
C-	Feedback	D-	Inhibitory pathway

9-	In case of low blood glucose concentration, the negative feedback is (MID 33)?		
A-	Conversion of glucose into glycogen	B-	Conversion of glucose into fat
C-	Conversion of glycogen into glucose	D-	Increase arterial blood pressure

10-	all of the following are negative feedback except: (MID 34)?
A-	activation of blood clotting factors

11-	Homeostasis : (MID 37)?
A-	Done by respiratory system only
B-	consumes no energy
C-	Is keeping the external surrounding environment normal
D-	Important for optimum cellular function

12-	About positive feedback, the true is … (FINAL 37)?
A-	The response of control system in the same direction of error
B-	The response of control system in the opposite direction of error

13-	in the homeostatic control system if the response is in the opposite direction of the initial change, this is considered: (Formative 36)?		
A-	Negative feedforward	B-	Positive Feedback
C-	Negative feedback	D-	Positive feedforward

14-	Which process maintains the internal environment? (2ND TRIAL 37)						
A-	Homeostasis	B-	Hemostasis	C-	Apoptosis	D-	Pinocytosis

15-	Which of the following receives messages and coordinates the response? (2ND TRIAL 37)						
A-	Control center	B-	Sensor	C-	Effector	D-	Receptor

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
A	C	A	A	A	A	C	C	C	A
11	12	13	14	15	16	17	18	19	20
D	A	C	A	A	-	-	-	-	-

Department Qs

1-	keeping the internal body environment constant is the ……: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	Haemostasis	B-	Homeostasis	C-	Haemolysis	D-	Feedback

2-	the internal body environment refers to: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	ICF	B-	ECG	C-	ECF	D-	Organs

3-	The internal environment: (SNU Bank, Dr.Mohamed Ashraf)?
A-	has Na+ as its major cation
B-	is the fluid found in cell
C-	normally has a pH of 7.0
D-	has a protein content greater than that of plasma

4-	Regarding the Homeostatic mechanism, which of the following detects the change and sends a message to the control center: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	Receptor	B-	stimulus	C-	effector	D-	response

5-	Regarding the Homeostatic mechanism, which of the following receives the message and coordinates a response: (SNU Bank, Dr.Mohamed Ashraf)?						
A-	control center	B-	effector	C-	response	D-	homeostasis

6-	Regarding the Homeostatic mechanism, which of the following responds to the commands from the control center (SNU Bank, Dr.Mohamed Ashraf)?						
A-	Stimulus	B-	Receptor	C-	Response	D-	Effector

7-	Regarding the Homeostatic mechanism, which of the following Increases or amplify the original change (SNU Bank, Dr.Mohamed Ashraf)?		
A-	negative feedback	B-	positive feedback
C-	involuntary control	D-	voluntary control

8-	Regarding the Homeostatic mechanism, which of the following tries to reverse or stop the original change (SNU Bank, Dr.Mohamed Ashraf)?		
A-	negative feedback	B-	positive feedback
C-	involuntary control	D-	Voluntary control

9-	regarding the regulation of body temperature, the receptors are (SNU Bank, Dr.Mohamed Ashraf)?
A-	the hypothalamic thermoreceptors
B-	the heat-regulating center in the hypothalamus of the brain
C-	The Sweat glands
D-	The blood vessels

10-	regarding the regulation of body temperature, the control center is ..... (SNU Bank, Dr.Mohamed Ashraf)?
A-	the thermoreceptors of the skin
B-	the heat-regulating center in the hypothalamus of the brain
C-	The Sweat glands
D-	The hypothalamic thermoreceptors

11-	Which of the following is an example of the negative feedback response of the homeostatic system: (SNU Bank, Dr.Mohamed Ashraf)?		
A-	Regulation of body temperature	B-	Regulation of Childbirth
C-	Regulation of the Blood clotting	D-	Regulation of ovulation

12-	Which of the following is an example of the positive feedback response of the homeostatic system: (SNU Bank, Dr.Mohamed Ashraf)?
A-	Regulation of body temperature
B-	Regulation of the oxygen and carbon dioxide levels in the blood
C-	Regulation of the Blood clotting
D-	Regulation of arterial blood pressure

13-	Which of the following systems doesn't contribute in homeostasis: (SNU Bank, Dr.Mohamed Ashraf)?				
A-	Circulatory system	B-	Blood	C-	Nervous system
D-	Musculoskeletal system	E-	None of the above		

14-	regarding the regulation of body temperature, the effectors are ..... (Dr. Mohamed Ashraf)?
A-	the thermoreceptors of the skin
B-	the heat-regulating center in the hypothalamus of the brain
C-	The Sweat glands
D-	The hypothalamic thermoreceptors

15-	ADH secretion increases when ..... increases: (Dr. Mohamed Ashraf)?		
A-	Blood volume	B-	Blood pressure
C-	Plasma osmolarity	D-	Sodium level

16-	ADH secretion increases when ..... decreases: (Dr. Mohamed Ashraf)?		
A-	Blood volume	B-	K level
C-	Plasma osmolarity	D-	Sodium level

17-	The receptors that detect blood osmolarity are the hypothalamic: (Dr. Mohamed Ashraf)?						
A-	Thermoreceptors	B-	Osmoreceptors	C-	Glucoreceptors	D-	Stretch receptors

18-	Which of the following has a bad effect on homeostatsis: (Dr. Mohamed Ashraf)?		
A-	Good nutrition	B-	Toxins
C-	Adequate rest, sunlight, and exercises	D-	All medications
E-	All psychological factors		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	C	A	A	A	D	B	A	A	B
11	12	13	14	15	16	17	18	19	20
A	C	E	C	C	A	B	B	-	-

M3WAN Bank

1-	The maintenance of a constant internal body environment for optimum cellular function is called: (Important M3WAN - Core)?						
A-	Homeostasis	B-	Osmosis	C-	Cell transport	D-	Endocytosis

2-	The "internal environment" in the concept of homeostasis refers to: (M3WAN - Core)?		
A-	Intracellular fluid only	B-	Transcellular fluid only
C-	The ECF and blood that bathe the cells	D-	Plasma only

3-	If a disturbance of homeostasis is not compensated, the ultimate result is: (M3WAN - Core)?		
A-	Permanent adaptation with no consequence	B-	Immediate restoration of the set point
C-	Enhanced cellular function	D-	Cellular dysfunction or even death

4-	Homeostasis is best described as: (M3WAN - Core)?
A-	A passive process requiring no energy
B-	An active process to which all body systems contribute
C-	A function limited to the nervous system alone
D-	A process limited to the endocrine system alone

5-	Continuous uptake of oxygen and nutrients by tissues from the interstitial fluid is compensated by continuous supply from: (M3WAN - Core)?		
A-	The lungs and the gastrointestinal tract	B-	The kidneys and the skin
C-	The liver and the spleen	D-	The heart and the blood vessels

6-	**Accumulation of CO2 and waste products in the interstitial fluid is prevented mainly by their continuous removal via: (M3WAN - Core)?**
A-	The gastrointestinal tract only
B-	The skeletal muscles
C-	The lungs and the kidneys/other excretory organs
D-	The bone marrow

7-	**Body temperature is kept constant at about 37°C mainly to: (M3WAN - Core)?**
A-	Reduce blood osmolarity
B-	Increase carbon dioxide production
C-	Prevent uterine contractions
D-	Preserve enzymatic activity and cellular function

8-	**Regulation of blood pH is maintained by all of the following EXCEPT: (M3WAN - Hard)?**						
A-	Buffer systems	B-	Skeletal muscle	C-	Respiration	D-	Kidney

9-	**The osmolarity of interstitial fluid and blood is normally kept constant at approximately: (M3WAN - Hard)?**		
A-	100-150 mosmol/L	B-	150-200 mosmol/L
C-	200-250 mosmol/L	D-	290-300 mosmol/L

10-	**Long-term regulation of blood pH is mainly maintained by the: (M3WAN - Core)?**						
A-	Kidney	B-	Liver	C-	Hypothalamus	D-	Adipose tissue

11-	**The locomotor system contributes to homeostasis mainly by: (M3WAN - Hard)?**		
A-	Directly regulating blood pH	B-	Synthesizing hormones
C-	Moving the person to the site of food	D-	Filtering waste products from blood

12-	**Regulation of the ionic composition of body fluids is partly achieved through hormonal secretions such as: (M3WAN - Core)?**		
A-	Insulin and glucagon	B-	Aldosterone and parathormone
C-	Acetylcholine and adrenaline	D-	Thyroxine and cortisol

13-	**In a homeostatic control system, the component that detects a change in the regulated variable is the: (M3WAN - Core)?**						
A-	Effector	B-	Efferent pathway	C-	Sensor	D-	Control center

14-	The pathway that carries information detected by a sensor to the control center in a homeostatic system is called the: (M3WAN - Core)?						
A-	Afferent pathway	B-	Efferent pathway	C-	Effector pathway	D-	Reflex arc

15-	The component of a homeostatic control system that integrates information and sends corrective signals to effectors is the: (M3WAN - Core)?						
A-	Sensor	B-	Afferent pathway	C-	Effector	D-	Control center

16-	The pathway that carries corrective signals from the control center to the effector in a homeostatic system is called the: (M3WAN - Core)?		
A-	Afferent pathway	B-	Efferent pathway
C-	Sensory pathway	D-	Receptor pathway

17-	Effectors in a homeostatic control system are best described as structures that: (M3WAN - Core)?
A-	Detect the initial change in the variable
B-	Transmit sensory information to the control center
C-	Integrate multiple afferent signals
D-	Respond to signals from the control center to correct the error

18-	Negative feedback is defined as a response that is: (Important M3WAN - Core)?
A-	In the same direction as the initiating change
B-	In the opposite direction to the initiating change
C-	Independent of the initiating change
D-	Restricted to endocrine systems only

19-	Increased sweating in response to a rise in body temperature is an example of: (M3WAN - Core)?		
A-	Positive feedback	B-	A homeostatic disturbance
C-	Negative feedback	D-	Transcellular regulation

20-	All of the following are examples of body homeostasis EXCEPT: (Important M3WAN - Hard)?
A-	Maintaining body temperature within a narrow range
B-	Regulation of blood glucose level
C-	The body making you thirsty when dehydrated
D-	Thinking

21-	Positive feedback is defined as a response that: (Important M3WAN - Core)?
A-	Moves in the same direction as the initiating change
B-	Opposes the initiating change

C-	Restores the variable to its original set point
D-	Is the main mechanism of homeostatic regulation

22-	**Which of the following is an example of positive feedback? (M3WAN - Core)?**		
A-	Regulation of blood glucose level	B-	Uterine contractions during labor
C-	Regulation of RBC number	D-	Regulation of body temperature

23-	**Regulation of the number of red blood cells is an example of: (M3WAN - Core)?**		
A-	Positive feedback	B-	Transcellular transport
C-	Negative feedback	D-	Osmotic regulation

24-	**Unlike negative feedback, positive feedback: (M3WAN - Hard)?**
A-	Returns the variable toward its original set point
B-	Is the predominant homeostatic mechanism in the body
C-	Requires no effector organ
D-	Continues to amplify the change until a specific end-point is reached

Answers - M3WAN Bank

1	2	3	4	5	6	7	8	9	10
A	C	D	B	A	C	D	B	D	A
11	12	13	14	15	16	17	18	19	20
C	B	C	A	D	B	D	B	C	D
21	22	23	24	25	26	27	28	29	30
A	B	C	D	-	-	-	-	-	-

إن النبي ﷺ شرع لأمته إيجاب إنكار المنكر؛ ليحصل بإنكاره من المعروف ما يحبُّه الله ورسوله ﷺ فإذا كان إنكار المنكر يستلزم ما هو أنكرُ منه، وأبغضُ إلى الله ورسوله ﷺ فإنه لا يسوغ إنكارُه، وإن كان الله يُبغضه ويَمقت أهلَه، وهذا كالإنكار على الملوك والولاة بالخروج عليهم؛ فإنه أساس كلِّ شر وفتنة إلى آخر الدهر، وقد استأذن الصحابة رسولَ الله ﷺ في قتال الأمراء الذين يؤخّرون الصلاة عن وقتها، وقالوا: أفلا نقاتلهم؟ فقال: «لا، ما أقاموا الصلاة»، وقال ﷺ: «مَن رأى من أميره ما يكرهه فليصبرْ، ولا ينزعنَّ يداً من طاعة».

• ابن القيم | أعلام الموقعين (٤/٣)

Functions of various cell organelles and cell membrane

Department Qs

1-	The functional and structural unit of the body is the: (Dr. Mohamed Ashraf)?						
A-	System	B-	Organ	C-	Tissue	D-	Cell

2-	The fluid portion of the cytoplasm is: (Dr. Mohamed Ashraf)?						
A-	Cytosol	B-	Ribosome	C-	Lysosome	D-	Centriole

3-	Gene expression and DNA replication is controlled by: (Dr. Mohamed Ashraf)?						
A-	Mitochondria	B-	Nucleus	C-	Golgi apparatus	D-	Ribosomes

4-	synthesis of proteins inside the cells occurs at: (Dr. Mohamed Ashraf)?						
A-	ribosomes	B-	lysosomes	C-	peroxisomes	D-	Desmosomes

5-	lipid synthesis and carbohydrate metabolism inside the cells occurs in: (Dr. Mohamed Ashraf)?		
A-	smooth endoplasmic reticulum	B-	Rough endoplasmic reticulum
C-	Golgi apparatus	D-	Nucleus

6-	Responsible for storing, modifying, and packaging of proteins into vesicles: (Dr. Mohamed Ashraf)?		
A-	smooth endoplasmic reticulum	B-	Rough endoplasmic reticulum
C-	Golgi apparatus	D-	Nucleus

7-	the organelle that contains oxidative enzymes such as catalase and they are a major site of oxygen utilization: (Dr. Mohamed Ashraf)?		
A-	ribosomes	B-	endoplasmic reticulum
C-	Peroxisomes	D-	Mitochondria

8-	the organelle that contains hydrolytic enzymes that can breakdown large molecules is: (Dr. Mohamed Ashraf)?						
A-	ribosomes	B-	lysosomes	C-	Peroxisomes	D-	Mitochondria

9-	Lysosomes are abundant in: (Dr. Mohamed Ashraf)?		
A-	Protein-secreting cells	B-	Neutrophils
C-	Nerve cells	D-	Muscle cells

10-	the major site of energy production inside the cells (powerhouse of the cell) is: (Dr. Mohamed Ashraf)?						
A-	nucleus	B-	Golgi apparatus	C-	Peroxisomes	D-	Mitochondria

11-	The smallest one of the components of the cytoskeleton is: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

12-	Which of the following is NOT a component of the cytoskeleton: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

13-	Which of the following components of the cytoskeleton is prominent where the cells are subjected to mechanical stress: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

14-	Motility of cilia or flagella is a function of: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

15-	Chromosome movement during cell division is a function of: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

16-	Transport of vesicles inside the cell is a function of: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

17-	Which of the following is found in the microvilli of the brush border of the intestine: (Dr. Mohamed Ashraf)?		
A-	Microtubules	B-	Microfilaments
C-	Intermediate filaments	D-	Myofibrils

18-	The cell membrane contains: (Dr. Mohamed Ashraf)?
A-	peripheral proteins which may function as enzymes or hormone receptors
B-	integral proteins which are necessary for the transport of respiratory gases
C-	channels through which alcohol can diffuse easily
D-	two layers of proteins

19-	The lipid bilayer of the cell membrane: (Dr. Mohamed Ashraf)?
A-	is composed entirely of phospholipids
B-	allows free diffusion of oxygen and alcohol through the membrane
C-	is highly permeable to $Na
D-	is permeable to $K

20-	Among functions of proteins in the cell membrane: (Dr. Mohamed Ashraf)?
A-	to carry metals and hormones
B-	to act as receptors for hormones
C-	to allow diffusion from extracellular space to inside cells against concentration and electrical gradient
D-	integral proteins are needed for the active transport of glucose into cells

21-	The cell membrane contains: (Dr. Mohamed Ashraf)?
A-	peripheral proteins to which hormones can bind to produce their effects
B-	lipids that prevent the diffusion of respiratory gases between extracellular and intracellular fluid
C-	integral proteins which are important for the transport of fat-soluble molecules
D-	an energy-dependent transport process by which K^{+} ions are normally removed from the cell

22-	Regarding the cell membrane are all true, EXCEPT: (Dr. Mohamed Ashraf)?		
A-	has integral proteins which form channels	B-	has a monolipid layer
C-	has peripheral proteins that act as receptors	D-	allow diffusion of O2 and CO2

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	A	B	A	A	C	C	B	B	D
11	12	13	14	15	16	17	18	19	20
B	D	C	A	A	A	B	A	B	B
21	22	23	24	25	26	27	28	29	30
A	B	-	-	-	-	-	-	-	-

M3WAN Bank

1-	The thickness of the cell membrane ranges from: (M3WAN - Hard)?						
A-	10 to 25 Å	B-	200 to 300 Å	C-	500 to 700 Å	D-	75 to 111 Å

2-	The cell membrane is also known as the: (M3WAN - Core)?		
A-	Sarcoplasmic reticulum	B-	Basal lamina

C-	Plasma membrane or plasmalemma	D-	Glycocalyx

3-	**Which of the following cells is an exception to the general rule that cells reproduce by division? (M3WAN - Core)?**						
A-	Hepatocyte	B-	Neuron	C-	Epithelial cell	D-	Fibroblast

4-	**Proteins constitute approximately what percentage of the cell membrane? (M3WAN - Core)?**						
A-	55%	B-	5%	C-	40%	D-	100%

5-	**Lipids constitute approximately what percentage of the cell membrane? (M3WAN - Core)?**						
A-	55%	B-	5%	C-	40%	D-	75%

6-	**The major lipids of the cell membrane are: (M3WAN - Core)?**		
A-	Phospholipids and glycolipids only	B-	Cholesterol and glycoproteins
C-	Triglycerides and phospholipids	D-	Phospholipids and cholesterol

7-	**In the lipid bilayer of the cell membrane, the hydrophobic tail portions of phospholipid molecules: (M3WAN - Core)?**		
A-	Face the extracellular fluid	B-	Meet in the center of the membrane
C-	Face the intracellular fluid only	D-	Bind directly to integral proteins

8-	**Cholesterol molecules arranged between phospholipid molecules mainly function to: (M3WAN - Core)?**
A-	Provide structural integrity by packing the phospholipids
B-	Form ion channels
C-	Act as membrane receptors
D-	Transport glucose across the membrane

9-	**The lipid layer of the cell membrane, being a semipermeable barrier, allows free passage of: (Important M3WAN - Core)?**
A-	Water-soluble substances such as glucose
B-	Charged electrolytes such as sodium and potassium
C-	Large water-soluble proteins
D-	Fat-soluble substances such as oxygen and carbon dioxide

10-	**Proteins that pass through the entire thickness of the cell membrane and are tightly bound to it are called: (M3WAN - Core)?**		
A-	Peripheral proteins	B-	Cytoskeletal proteins
C-	Integral (transmembrane) proteins	D-	Glycoproteins only

11-	**Which of the following is an example of an integral membrane protein? (M3WAN - Hard)?**
A-	Cytoskeletal protein
B-	Channel protein
C-	A protein that dissociates readily from the membrane
D-	A protein confined to the inner surface only, never spanning the membrane

12-	**Peripheral membrane proteins differ from integral proteins in that they: (M3WAN - Core)?**
A-	Are loosely bound and dissociate readily from the membrane
B-	Span the entire thickness of the membrane
C-	Are tightly bound transmembrane structures
D-	Always function as ion channels

13-	**Channel proteins in the cell membrane mainly help in the diffusion of: (M3WAN - Core)?**
A-	Fat-soluble gases such as O2 and CO2
B-	Steroid hormones
C-	Cholesterol
D-	Water-soluble substances such as glucose and electrolytes

14-	**Carrier (transport) proteins in the cell membrane function to: (M3WAN - Core)?**
A-	Provide the structural framework of the cytoskeleton
B-	Store genetic information
C-	Transport substances across the cell membrane by active or passive transport
D-	Synthesize ATP directly

15-	**Some carrier proteins act as pumps, meaning they: (M3WAN - Core)?**
A-	Actively transport ions across the cell membrane
B-	Passively allow ions to diffuse down their gradient
C-	Store hormones for later release
D-	Break down worn-out organelles

16-	**Receptor proteins in the cell membrane mainly serve as binding sites for: (M3WAN - Core)?**		
A-	Structural lipids only	B-	Cytoskeletal filaments
C-	Genetic material	D-	Hormones and neurotransmitters

17-	**Some cell membrane proteins function as enzymes that: (M3WAN - Hard)?**
A-	Store hereditary information
B-	Provide the hydrophobic core of the membrane

C-	Form the glycocalyx exclusively
D-	Control chemical (metabolic) reactions within the membrane

18-	**Some cell membrane proteins act as antigens, meaning they: (M3WAN - Core)?**		
A-	Directly synthesize antibodies	B-	Prevent immune recognition of the cell
C-	Induce the process of antibody formation	D-	Transport lipids only

19-	**Cell adhesion proteins (integral proteins) are mainly responsible for: (M3WAN - Core)?**
A-	Synthesis of membrane lipids
B-	Attachment of cells to their neighbors or to the basal lamina
C-	Generation of the resting membrane potential
D-	Detoxification of metabolic waste

20-	**Functions of cell membrane proteins include all of the following EXCEPT: (Important M3WAN - Hard)?**
A-	Acting as a barrier to water-soluble substances
B-	Acting as carriers for active or passive transport
C-	Serving as receptor sites for hormones
D-	Forming enzymes that control membrane reactions

21-	**The thin, loose carbohydrate covering over the entire surface of the cell membrane is called the: (M3WAN - Core)?**						
A-	Cytoskeleton	B-	Sarcolemma	C-	Basal lamina	D-	Glycocalyx

22-	**A major function of carbohydrate molecules in the cell membrane is to: (Important M3WAN - Core)?**
A-	Provide the main pathway for active transport
B-	Form the hydrophobic core of the membrane
C-	Serve as receptors and mediate cell-to-cell communication
D-	Provide mechanical strength to the membrane

23-	**The glycocalyx contributes to tissue architecture mainly by: (M3WAN - Core)?**
A-	Forming the hydrophobic core of the membrane
B-	Helping in the tight fixation of neighboring cells to one another
C-	Actively pumping ions across the membrane
D-	Generating action potentials

24-	**Protection of the cytoplasm and the organelles within it is a function of the cell membrane known as its: (M3WAN - Core)?**		
A-	Protective function	B-	Absorptive function

C-	Excretory function	D-	Exchange of gases function

25-	**The function of the cell membrane that allows only certain substances to pass while barring others is called: (M3WAN - Core)?**		
A-	Absorptive function	B-	Excretory function
C-	Protective function	D-	Selective permeability

26-	**Absorption of nutrients into the cell through the cell membrane is an example of its: (M3WAN - Core)?**		
A-	Protective function	B-	Excretory function
C-	Absorptive function	D-	Maintenance of shape function

27-	**Excretion of metabolites and other waste products from the cell through the cell membrane is an example of its: (M3WAN - Core)?**		
A-	Absorptive function	B-	Excretory function
C-	Protective function	D-	Exchange of gases function

28-	**Entry of oxygen into the cell and exit of carbon dioxide through the cell membrane is an example of its: (M3WAN - Core)?**		
A-	Exchange of gases function	B-	Absorptive function
C-	Excretory function	D-	Protective function

29-	**Maintenance of the shape and size of the cell is a function attributed mainly to the: (M3WAN - Core)?**						
A-	Nucleus	B-	Golgi apparatus	C-	Peroxisome	D-	Cell membrane

30-	**Which of the following cytoplasmic organelles does NOT have a limiting membrane? (Important M3WAN - Hard)?**						
A-	Mitochondria	B-	Lysosome	C-	Ribosome	D-	Golgi apparatus

31-	**Which of the following cytoplasmic organelles HAS a limiting membrane? (M3WAN - Core)?**		
A-	Ribosome	B-	Mitochondria
C-	Cytoskeletal filament	D-	Microtubule

32-	**The rough endoplasmic reticulum is mainly responsible for: (M3WAN - Core)?**		
A-	Synthesis of proteins	B-	Synthesis of lipids and steroids
C-	Storage and metabolism of calcium	D-	Detoxification of hydrogen peroxide

33-	**Besides protein synthesis, the rough endoplasmic reticulum also functions in: (M3WAN - Hard)?**		
A-	Storage and metabolism of calcium	B-	Synthesis of lipids and steroids
C-	Detoxification of hydrogen peroxide	D-	Degradation of worn-out organelles

34-	The smooth endoplasmic reticulum is responsible for all of the following EXCEPT: (M3WAN - Core)?
A-	Synthesis of lipids and steroids
B-	Storage and metabolism of calcium
C-	Synthesis of proteins
D-	Catabolism and detoxification of toxic substances

35-	Processing, packaging, labeling, and delivery of proteins and lipids is mainly the function of the: (M3WAN - Core)?		
A-	Golgi apparatus	B-	Rough endoplasmic reticulum
C-	Peroxisome	D-	Centrosome

36-	Degradation of macromolecules and worn-out organelles is mainly carried out by the: (M3WAN - Core)?						
A-	Golgi apparatus	B-	Lysosome	C-	Mitochondria	D-	Ribosome

37-	In addition to degradation of macromolecules, lysosomes are also responsible for: (M3WAN - Hard)?
A-	Synthesis of ATP
B-	Synthesis of RNA
C-	Formation of ribosomal subunits
D-	Secretion of perforin, granzymes, melanin, and serotonin

38-	Detoxification of hydrogen peroxide is mainly a function of the: (M3WAN - Hard)?						
A-	Lysosome	B-	Mitochondria	C-	Peroxisome	D-	Nucleus

39-	Functions of peroxisomes include all of the following EXCEPT: (Important M3WAN - Hard)?		
A-	Breakdown of excess fatty acids	B-	Synthesis of proteins
C-	Degradation of purine to uric acid	D-	Role in the formation of myelin and bile acids

40-	Movement of chromosomes during cell division is mainly the function of the: (M3WAN - Hard)?						
A-	Centrosome	B-	Nucleus	C-	Cytoskeleton	D-	Ribosome

41-	Production of energy and synthesis of ATP is mainly the function of the: (M3WAN - Core)?						
A-	Ribosome	B-	Nucleus	C-	Cytoskeleton	D-	Mitochondria

42-	Synthesis of proteins is the main function of the: (M3WAN - Core)?						
A-	Golgi apparatus	B-	Peroxisome	C-	Ribosome	D-	Centrosome

43-	Determination and stability of cell shape, together with cellular movement, are functions of the: (M3WAN - Core)?						
A-	Nucleus	B-	Cytoskeleton	C-	Mitochondria	D-	Lysosome

44-	Control of all cellular activities and synthesis of RNA is mainly the function of the: (M3WAN - Core)?						
A-	Mitochondria	B-	Cytoskeleton	C-	Golgi apparatus	D-	Nucleus

45-	Formation of the subunits of ribosomes is mainly a function of the: (M3WAN - Hard)?		
A-	Nucleus	B-	Golgi apparatus
C-	Rough endoplasmic reticulum	D-	Mitochondria

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	C	B	A	C	D	B	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	D	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	C	B	A	D	C	B	A	D	C
31	32	33	34	35	36	37	38	39	40
B	A	D	C	A	B	D	C	B	A
41	42	43	44	45	46	47	48	49	50
D	C	B	D	A	-	-	-	-	-

«عِلمُ الصحابةِ رضوان الله عليهم»

كان الصّحابةُ أعرفَ الأُمّةِ بالإسلام وتَفاصيله وأبوابِه وطُرقه، وأشدَّ النّاسِ رغبةً فيه، ومحبّةً له، وجهادًا لأعدائه، وتكلُّمًا بأعلامه، وتَحذيرًا من خلافه؛ لكمالِ علمهم بضدِّه

•ابن القيم | مفتاح دار السعادة (٢/٨٣٧)

Transport across the cell membrane

Pre Exams & Formative

1-	**Which properties are characteristic of ion channels (MID 31)?**
A-	They are usually lipids
B-	They are always open
C-	They mediate simple diffusion
D-	They exist on one side of the plasma membrane

2-	**If a small amount of small sized lipid soluble non charged particles were added to an isotonic saline solution containing cells, what would be the result (MID 31)?**		
A-	The cells would shrink	B-	The cells would change their shape
C-	The cells would swell	D-	The cells maintain their size

3-	**Which of the following does not directly or indirectly require an energy source (MID 31)?**		
A-	Primary active transport	B-	Facilitated diffusion of glucose
C-	The Na+/K+ ATPase pump	D-	Uphill transport

4-	**Regarding diffusion, which is correct (MID 31)?**
A-	Diffusion of a solute through a membrane is considerably quicker than diffusion of the same solute through a water layer
B-	A single type ion, such as Na+, can diffuse through one type of channel only
C-	Water-soluble solutes diffuse more readily through a plasma membrane than do lipid-soluble ones
D-	The rate of facilitated diffusion of a solute is determined by the number of transporters in the membrane

5-	**Plasma has an osmolarity of about 300 mOsm. The osmolarity of hypertonic solution is equal to (MID 31)?**						
A-	150 mOsm	B-	300 mOsm	C-	250 mOs	D-	300 Osm

6-	**In considering diffusion of ions through an ion channel, the driving force may include (MID 31, FINAL 31, FINAL 34)?**		
A-	The electrical gradient	B-	Osmosis Process
C-	ATP availability	D-	Secondary active transport

7-	**Which of these statements about the facilitated diffusion of glucose is TRUE (MID 31)?**
A-	There is a net movement from the region of lower to the region of higher concentration
B-	Carrier proteins in the cell membrane are required for this transport

C-	This transport requires energy obtained from ATP
D-	It is an example 2nd active transport

8-	**Which of these statements about the Na+/K+ pump is TRUE (MID 31)?**		
A-	Na+ is actively transported into the cell	B-	K+ is actively transported out of the cell
C-	More K+ ions are moved if compared to Na	D-	More Na+ ions are moved if compared to K

9-	**Which of the following solutions is hypertonic (MID 31)?**						
A-	0.9 % NaCl	B-	20% mannitol	C-	5%dextrose	D-	Distilled water

10-	**Red blood cells from a healthy individual Were placed in each of the following solutions and Observations were made after l hour. Cells would have most likely have lysed when placed in (MID 31)?**						
A-	0.3%NaCl	B-	0.9%NaCl	C-	1.2%NaCl	D-	Distilled water

11-	**An isotonic solution (FINAL 31)?**
A-	Has a concentration of about 0.9% NaCl
B-	Rarely exists within the natural body fluids
C-	Results in a net osmosis of water from the blood plasma into the RBCs
D-	Result in RBCs decrease in size

12-	**Which properties are characteristic of ion channels (FINAL 31)?**
A-	They are always closed
B-	They permit movement of ions against electrochemical gradients
C-	They mediate facilitated diffusion
D-	They are protein in nature

13-	**Which of the following directly or indirectly require an energy source (FINAL 31, FINAL 34)?**		
A-	Secondary active transport	B-	Downhill transport
C-	Leak	D-	facilitated diffusion of glucose

14-	**Which of these statements about the Na^+/K^+ pump is FALSE (FINAL 31)?**
A-	The pumps are present in all cells
B-	Na^+ is actively transported out of the cell
C-	K^+ is actively transported into the cell
D-	An equal number of Na^+ and K^+ are transported with each cycle of the pump

15-	**A solution having a greater conc. of water than in the cell is said to be (FINAL 31)?**						
A-	Hypertonic	B-	Hypotonic	C-	Isotonic	D-	Heterotonic

16-	A postoperative patient is ordered an I.V. solution of dextrose 5%in normal saline solution. What type of fluid is this solution (MID 32)?						
A-	Hypertonic	B-	Hypotonic	C-	Isotonic	D-	Colloid

17-	Which of the following methods does not require any carrier or channel for transport of substances (MID 32)?		
A-	secondary active transport	B-	facilitated diffusion
C-	simple diffusion	D-	primary active transport

18-	Which substances are transported through facilitated diffusion (MID 32)?						
A-	glucose	B-	urea	C-	oxygen	D-	alcohol

19-	Which type of movement occurs when Na/K pump is used (MID 32)?		
A-	Na ions moves out of the cell and K+ move in	B-	K+ ion moves out of cell and Na ion move in
C-	both Na and K+ ions move inside the cell	D-	both Na and K+ move out of the cell

20-	A cell stop ATP production, which of the following transport processes would stop (FINAL 31)?		
A-	The movement of Na^+ out of a cell	B-	The movement of Na^+ into of a cell
C-	The movement of water into cell	D-	The movement of K^+ out of a cell

21-	A fireman sustains burns while fighting an apartment fire. He receives IV fluid. Which type of fluid is used (FINAL 32)?		
A-	Normal saline solution	B-	Half-normal saline solution
C-	Lactated Ringer's solution	D-	Dextrose 5% in lactated Ringer's solution

22-	When a hypotonic crystalloid ation is infused into the bloodstream, causes the cells to (FINAL 32)?						
A-	shrink	B-	swell	C-	release chloride	D-	release potassium

23-	Which substance is transported through secondary active transport (FINAL 32)?						
A-	urea	B-	water	C-	O2	D-	none of the above

24-	What is the source of energy in active transport (FINAL 32)?		
A-	from sunlight	B-	hydrolysis of ATP
C-	energy stored in ionic concentration gradient	D-	both A and C

25-	Which of the following statement is NOT true (FINAL 32)?
A-	chemical changes leads to change in shape of pump
B-	symporters are the one which moves substances in different direction

C-	Vitamin K is transported through simple diffusion
D-	in passive transport, the substance moves down its concentration gradient

26-	**Facilitated diffusion differs from simple diffusion in that (MID 33)?**
A-	It requires no carrier
B-	Itmoves substances from high to low concentration
C-	It requires energy
D-	It is a faster rate of transport

27-	**Secondary active transport and facilitated diffusion are similar in that they (MID 33)?**						
A-	Use energy	B-	Use carriers	C-	Transport water	D-	Use channels

28-	**Symport carrier (MID 33, 2ND TRIAL 35)?**
A-	transports only one substance
B-	transport substances in opposite directions
C-	transport two or more substances in the same direction
D-	Is not an integral protein

29-	**what is the type of transport in Na voltage gated channels in active transport (MID 34)?**				
A-	Facilitated diffusion	B-	Simple diffusion	C-	Primary active transport

30-	**Which of the following doesn't exhibit saturation kinetics (MID 34)?**
A-	Potassium pass through cell membrane through leak potassium channel

31-	**which of the following is primary active transport (MID 34)?**
A-	Ca ATPase

32-	**ions iflux through cell membrane occurs easily in (MID 34)?**
A-	simple diffusion

33-	**Which type of transport is Na-glucose transported from intestine (FINAL 34)?**
A-	secondary active transport

34-	**Blockage of Na/K ATP based pump would lead to (FINAL 34)?**		
A-	Extracellular edema	B-	Intracellular edema
C-	nonpitting edema	D-	no effect

35-	Transports Na and K in cell by (2ND TRIAL 35)?						
A-	Uniporter	B-	Symporter	C-	Antiporter	D-	Diporter

36-	Rate of diffusion is increased by (2ND TRIAL 35)?				
A-	Increase pressure gradient	B-	Increase body temperature	C-	A & B

37-	What is the best solution for postoperative hypotension (FINAL 36)?		
A-	Ringer's lactate	B-	0.9% NaCl
C-	Glucose / dextrose	D-	None of the above

38-	Passive transport through cell membrane has the following characters except: (MID 37)?		
A-	Can occur through lipid bilayer	B-	Can occur through proteins channels
C-	may be Carrier mediated	D-	ATP dependent

39-	Types of gated channels include all the following except: (MID 37)?		
A-	Voltage channels	B-	leak channels
C-	Ligand channels	D-	Mechanical channels

40-	Transport of glucose proceeds down an electrochemical gradient by which of the following processes? (Formative 36)?		
A-	Facilitated diffusion	B-	Simple diffusion
C-	Cotransport	D-	Primary active transport

41-	In secondary active transport, Na+ ions transports: (Formative 37)?		
A-	against its concentration gradient	B-	directly hydrolysis metabolic energy
C-	with its concentration gradient	D-	Requires uniport carrier

42-	Which of the following is shared by simple and facilitated diffusion? (2ND TRIAL 37)
A-	Require a carrier protein
B-	Require energy
C-	Saturable kinetics
D-	Run down the concentration gradient

43-	Which is true about primary active transport? (2ND TRIAL 37)
A-	Sodium ions pass through ion channels by it
B-	Transports substances down their concentration gradient
C-	Requires energy
D-	Does not utilize ATP

44-	Passage of Na^+ ions through a leak channel occurs by: (2ND TRIAL 37)
A-	Facilitated diffusion
B-	Simple diffusion
C-	Primary active transport
D-	Secondary active transport

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
C	C	B	D	D	A	B	D	B	D
11	12	13	14	15	16	17	18	19	20
A	D	A	D	B	A	C	A	A	A
21	22	23	24	25	26	27	28	29	30
C	B	D	B	B	D	B	C	A	A
31	32	33	34	35	36	37	38	39	40
A	A	A	B	C	C	A	D	B	A
41	42	43	44	45	46	47	48	49	50
C	D	C	B	-	-	-	-	-	-

Lecture Qs

1-	Which of the following characteristics is shared by simple and facilitated diffusion of glucose (Lecture Qs 37)?		
A-	Occurs down an electrochemical gradient	B-	Is saturable
C-	Requires metabolic energy	D-	Is inhibited by the presence of galactose
E-	Requires a Na+ gradient		

2-	The most abundant cation in ICF is (Lecture Qs 37)?						
A-	Sodium	B-	Potassium	C-	Magnesium	D-	Calcium

3-	The most abundant anion ECF is (Lecture Qs 37)?						
A-	bicarbonate	B-	chloride	C-	phosphate	D-	protein anion

4-	A new drug is developed that blocks the transporter for H+ secretion in gastric parietal cells. Which of the following transport processes is being inhibited (Lecture Qs 37)?		
A-	Simple diffusion	B-	Facilitated diffusion
C-	Primary active transport	D-	Cotransport
E-	Counter transport		

5-	Which of the following transport processes is mediated by a carrier protein in the plasma membrane? (Select all correct answers). (Lecture Qs 37)?		
A-	Glucose uptake through SGLT-1	B-	Na influx through Na channels
C-	Na-K ATPase	D-	Water fluxes through aquaporins

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	B	B	C	A&C	-	-	-	-	-

Department Qs

1-	the type of transport that doesn't need a carrier or energy is (Dr. Mohamed Ashraf)?		
A-	Simple diffusion	B-	Facilitated diffusion
C-	Active transport	D-	Vesicular transport

2-	A lipid-soluble substance can pass the cell membrane through (Dr. Mohamed Ashraf)?		
A-	The phospholipid bilayer	B-	Carrier protein
C-	Protein channels	D-	Lipid carrier

3-	Oxygen is transported from blood to the cells by (Dr. Mohamed Ashraf)?		
A-	Simple diffusion	B-	Active transport
C-	Facilitated diffusion	D-	osmosis

4-	Sodium ions to pass through the cell membrane from high to low concentration, it needs (Dr. Mohamed Ashraf)?						
A-	A protein channel	B-	ATP	C-	A carrier protein	D-	ATP and carrier

5-	the type of transport that doesn't need energy but needs a carrier is: (Dr. Mohamed Ashraf)?		
A-	Simple diffusion	B-	Facilitated diffusion
C-	Active transport	D-	Vesicular transport

6-	Glucose is transported from blood to the cells by (Dr. Mohamed Ashraf)?		
A-	Simple diffusion	B-	Active transport
C-	Facilitated diffusion	D-	Osmosis

7-	Facilitated diffusion differs from simple diffusion in that (Dr. Mohamed Ashraf)?		
A-	Facilitated diffusion needs energy	B-	Simple diffusion needs energy

C-	Simple diffusion needs carrier	D-	Facilitated diffusion needs carrier

8-	**Facilitated diffusion: (Dr. Mohamed Ashraf)?**
A-	needs a constant supply of energy
B-	can work against electrochemical gradient
C-	stops when all the binding sites of the carrier protein are filled
D-	can work against concentration gradient

9-	**The rate of diffusion of a solute across a membrane depends on all, EXCEPT: (Dr. Mohamed Ashraf)?**
A-	total area of membrane exposed to the solute
B-	electrical gradient for the solute across the membrane
C-	ATP as a source of energy
D-	concentration gradient for the solute across the membrane

10-	**the rate of diffusion is affected by: (Dr. Mohamed Ashraf)?**		
A-	concentration gradient	B-	number of mitochondria
C-	amount of ATP in the cell	D-	blood pressure

11-	**the rate of diffusion of a substance is inversely proportionate with: (Dr. Mohamed Ashraf)?**		
A-	concentration gradient	B-	solubility of the substance
C-	temperature	D-	molecular weight of the substance

12-	**Diffusion of solutes across semi-permeable membrane will decrease if there is an increase in the: (Dr. Mohamed Ashraf)?**
A-	area of the membrane
B-	thickness of the membrane
C-	solute concentrate difference across the membrane
D-	temperature of the particle

13-	**The rate of diffusion of a particle across a membrane will increase if: (Dr. Mohamed Ashraf)?**
A-	the concentration gradient for the particle decreases
B-	the size of the particle increases
C-	the lipid solubility of the particle increases
D-	the area of the membrane decreases

14-	**Water pass from area of low solute to area of high solute concentration by (Dr. Mohamed Ashraf)?**		
A-	Simple diffusion	B-	Exocytosis
C-	Facilitated diffusion	D-	Osmosis

15-	**Concerning osmosis across selectively permeable membrane, all are true, EXCEPT: (Dr. Mohamed Ashraf)?**
A-	it is the movement of solute across a semi-permeable membrane from an area in which it is in low concentration to an area in which it is in high concentration
B-	it is caused by a concentration difference of water
C-	the osmotic pressure is the amount of pressure required to stop osmosis completely
D-	the osmotic pressure is determined by the number of particles per unit volume of fluid

16-	**Osmotic pressure across the capillary wall depends on: (Dr. Mohamed Ashraf)?**		
A-	Size of the molecule	B-	Shape of the molecule
C-	Concentration of the molecule	D-	All of the above

17-	**The solution which contains an electrolyte concentration similar to that in plasma is: (Dr. Mohamed Ashraf)?**		
A-	Isotonic solution	B-	Hypotonic solution
C-	Hypertonic solution	D-	Isometric solution

18-	**The solution which contains an electrolyte concentration higher than that in plasma is: (Dr. Mohamed Ashraf)?**		
A-	Isotonic solution	B-	Hypotonic solution
C-	Hypertonic solution	D-	Isometric solution

19-	**The solution which contains an electrolyte concentration lower than that in plasma is: (Dr. Mohamed Ashraf)?**		
A-	Isotonic solution	B-	Hypotonic solution
C-	Hypertonic solution	D-	Isometric solution

20-	**the type of transport that needs a carrier and energy is (Dr. Mohamed Ashraf)?**		
A-	Simple diffusion	B-	Facilitated diffusion
C-	Active transport	D-	Osmosis

21-	**an example of secondary active transport is (Dr. Mohamed Ashraf)?**		
A-	Na-K pump	B-	Na diffusion
C-	Na-glucose cotransport	D-	glucose diffusion

22-	**an example of Primary active transport is (Dr. Mohamed Ashraf)?**		
A-	Na-K pump	B-	Na diffusion
C-	Na-glucose cotransport	D-	glucose diffusion

23-	the carrier that transports one substance is called: (Dr. Mohamed Ashraf)?						
A-	uniporter	B-	Symporter	C-	Antiporter	D-	Diporter

24-	the carrier that transports two substances in the same direction is called: (Dr. Mohamed Ashraf)?						
A-	uniporter	B-	Symporter	C-	Antiporter	D-	Diporter

25-	the carrier that transports two substances in opposite directions is called: (Dr. Mohamed Ashraf)?						
A-	uniporter	B-	Symporter	C-	Antiporter	D-	Diporter

26-	Regarding transport across the cell membrane: (Dr. Mohamed Ashraf)?
A-	facilitated diffusion is an energy requiring process
B-	active transport is necessary for the movement of molecules against electrochemical gradient
C-	water soluble substances of large molecular size diffuse through the phospholipids
D-	only fat-soluble substances can pass easily through the pores created by the integral proteins

27-	Regarding the transport mechanisms across the cellular membrane: (Dr. Mohamed Ashraf)?
A-	facilitated diffusion is an active process
B-	ions pass freely through protein channels
C-	Na+/K+ pump is an example of secondary active transport
D-	counter transport is transporting 2 molecules in the same direction

28-	Which of the following statements regarding exocytosis is correct: (Dr. Mohamed Ashraf)?
A-	Is always employed by cells for secretion
B-	Takes up large molecules from the extracellular space
C-	Allows the retrieval of elements of the cell membrane
D-	Is used to deliver materials to the extracellular space

29-	Cell drinking is the: (Dr. Mohamed Ashraf)?						
A-	Exocytosis	B-	Phagocytosis	C-	Pinocytosis	D-	Osmosis

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
A	A	A	A	B	C	D	C	C	A
11	12	13	14	15	16	17	18	19	20
D	B	C	D	A	C	A	C	B	C
21	22	23	24	25	26	27	28	29	30
C	A	A	B	C	B	B	D	C	-

M3WAN Bank

1-	**Passive transport across the cell membrane is characterized by: (M3WAN - Core)?**
A-	Requiring hydrolysis of ATP
B-	Movement against the concentration gradient
C-	Requiring a carrier protein in all cases
D-	Movement along the concentration or electrical gradient without energy expenditure

2-	**Active transport differs from passive transport mainly in that active transport: (M3WAN - Core)?**
A-	Always moves substances along their concentration gradient
B-	Never requires a carrier protein
C-	Requires energy from ATP breakdown to move substances against their gradient
D-	Occurs only through channel proteins

3-	**According to the method of transport, transport across the cell membrane is classified into passage through the lipid bilayer, passage through membrane proteins, and: (M3WAN - Core)?**		
A-	Active transport	B-	Vesicular transport
C-	Passive transport	D-	Osmotic transport

4-	**Simple diffusion through the lipid bilayer is the suitable pathway for: (M3WAN - Core)?**
A-	Lipid-soluble substances such as O2, CO2, and fatty acids
B-	Water-soluble substances such as glucose
C-	Charged electrolytes such as Na+ and K
D-	Large molecular weight proteins

5-	**Substances transported through channel or carrier proteins include all of the following EXCEPT: (M3WAN - Hard)?**		
A-	Water	B-	Glucose and amino acids
C-	Lipid-soluble gases such as O2	D-	Electrolytes such as Na+ and K

6-	**Aquaporins are membrane channel proteins specifically designed for the passage of: (M3WAN - Core)?**						
A-	Glucose	B-	Sodium ions	C-	Amino acids	D-	Water

7-	**Ungated ion channels in the cell membrane are characterized by being: (M3WAN - Core)?**		
A-	Opened only by voltage change	B-	Continuously open
C-	Opened only by ligand binding	D-	Opened only by mechanical deformation

8-	**Opening of calcium channels at the axon terminal of the neuromuscular junction upon arrival of an action potential is an example of? (Important M3WAN - Core)**		
A-	Voltage-gated channel	B-	Ligand-gated channel
C-	Mechanically gated channel	D-	Ungated channel

9-	**Acetylcholine causing the opening of sodium channels in the postsynaptic membrane at the neuromuscular junction is an example of? (Important M3WAN - Core)**		
A-	Voltage-gated channel	B-	Mechanically gated channel
C-	Ungated channel	D-	Ligand-gated channel

10-	**Opening of sodium channels in a pressure receptor due to deformation of its core fiber under pressure is an example of? (M3WAN - Hard)**				
A-	Voltage-gated channel	B-	Ligand-gated channel	C-	Mechanically gated channel
D-	Ungated channel				

11-	**Facilitated diffusion differs from simple diffusion in that facilitated diffusion: (Important M3WAN - Core)?**
A-	Requires no carrier protein
B-	Occurs at a faster rate of transport with the help of a carrier protein
C-	Requires energy from ATP
D-	Moves substances against their concentration gradient

12-	**Glucose and amino acids are transported across the cell membrane mainly by: (M3WAN - Core)?**		
A-	Facilitated diffusion	B-	Simple diffusion through the lipid bilayer
C-	Primary active transport	D-	Osmosis

13-	**The rate of diffusion across the cell membrane is: (M3WAN - Core)?**
A-	Inversely proportional to body temperature
B-	Directly proportional to the size of the molecule
C-	Independent of the concentration gradient
D-	Inversely proportional to the thickness of the cell membrane

14-	**All of the following are types of diffusion across the cell membrane EXCEPT: (M3WAN - Core)?**		
A-	Simple diffusion through the lipid layer	B-	Facilitated diffusion
C-	Primary active transport	D-	Osmosis

15-	**Osmosis is defined as the movement of water through a semipermeable membrane from an area of: (M3WAN - Core)?**
A-	Lower solute concentration to higher solute concentration
B-	Higher solute concentration to lower solute concentration
C-	Lower solvent concentration to higher solvent concentration
D-	High hydrostatic pressure to low hydrostatic pressure

16-	**Osmotic pressure is best defined as the pressure required to: (M3WAN - Core)?**
A-	Force water out of a hypotonic solution
B-	Prevent osmotic flow of water into a solution with higher solute concentration
C-	Move solutes across a semipermeable membrane
D-	Equalize hydrostatic pressure across a capillary wall

17-	**Osmolarity of a solution refers to: (M3WAN - Core)?**
A-	Only the solutes that cannot penetrate the cell membrane
B-	Only the solutes that can penetrate the cell membrane
C-	The total volume of solvent in the solution
D-	The total number of solute particles in the solution, whether or not they can penetrate the cell membrane

18-	**Solutes that cannot penetrate the cell membrane are described as: (M3WAN - Hard)?**
A-	Osmotically useless
B-	Freely diffusible
C-	Osmotically active/useful, since they can cause osmosis
D-	Isotonic by definition

19-	**Solutes that CAN penetrate the cell membrane are described as: (M3WAN - Hard)?**
A-	Osmotically active
B-	Osmotically useless, since they cannot cause osmosis
C-	Osmotically useful
D-	Hypertonic by definition

20-	**Placing a cell in an isotonic solution results in: (M3WAN - Core)?**		
A-	No change in cell volume	B-	An increase in cell volume
C-	A decrease in cell volume	D-	Immediate cell lysis

21-	**Placing a cell in a hypotonic solution results in: (M3WAN - Core)?**
A-	A decrease in cell volume due to water leaving the cell
B-	No change in cell volume

C-	Immediate loss of membrane potential
D-	An increase in cell volume due to water entering the cell

22-	**Placing a cell in a hypertonic solution results in: (M3WAN - Core)?**
A-	An increase in cell volume due to water entering the cell
B-	No change in cell volume
C-	A decrease in cell volume due to water leaving the cell
D-	Immediate cell lysis

23-	**Which of the following solutions is hypertonic relative to intracellular fluid? (Important M3WAN - Core)?**						
A-	0.9% NaCl	B-	20% mannitol	C-	5% dextrose	D-	Distilled water

24-	**Red blood cells placed in which of the following solutions would most likely undergo lysis? (Important M3WAN - Core)?**						
A-	0.3% NaCl	B-	0.9% NaCl	C-	1.2% NaCl	D-	20% mannitol

25-	**Active transport of substances across the cell membrane always requires: (M3WAN - Core)?**		
A-	A channel protein only	B-	Simple diffusion through the lipid layer
C-	No membrane protein	D-	A carrier protein

26-	**Carrier proteins differ from channel proteins in that carrier proteins: (M3WAN - Hard)?**
A-	Form a continuous open pore across the membrane
B-	Never span the entire membrane thickness
C-	Have no lumen but change their orientation to move a substance across the membrane
D-	Function only in vesicular transport

27-	**Primary active transport is defined as the type of transport in which energy is: (M3WAN - Core)?**
A-	Derived indirectly from the sodium concentration gradient
B-	Liberated directly from the breakdown of ATP
C-	Derived from the potassium concentration gradient
D-	Not required at all

28-	**With each cycle of the sodium-potassium pump, per ATP molecule hydrolyzed: (Important M3WAN - Core)?**
A-	Three Na+ ions are pumped out and two K+ ions are pumped in
B-	Two Na+ ions are pumped out and three K+ ions are pumped in
C-	Three Na+ and three K+ ions are exchanged equally

D-	Two Na+ ions are pumped out and two K+ ions are pumped in

29-	**The Na+-K+ pump is described as electrogenic because it: (M3WAN - Core)?**
A-	Forces a net negative charge out of the cell
B-	Transports equal numbers of positive and negative charges
C-	Has no effect on membrane charge
D-	Forces a net positive charge out of the cell

30-	**A carrier protein that transports only one substance across the cell membrane is called: (M3WAN - Core)?**						
A-	Symport	B-	Antiport	C-	Uniport	D-	Aquaporin

31-	**A carrier protein that transports two or more substances from one side of the membrane to the other in the same direction is called: (Important M3WAN - Core)?**		
A-	Uniport	B-	Symport (co-transport)
C-	Antiport (counter-transport)	D-	Ungated channel

32-	**A carrier protein that transports one substance into the cell while simultaneously transporting another substance out of the cell is called: (M3WAN - Core)?**		
A-	Antiport (counter-transport)	B-	Uniport
C-	Symport (co-transport)	D-	Aquaporin

33-	**Competitive inhibition in carrier-mediated transport occurs when: (M3WAN - Core)?**
A-	A carrier protein becomes saturated by a single substance
B-	Two dissimilar molecules use the same carrier with two different binding sites
C-	ATP is unavailable for transport
D-	Two similar molecules compete for the same binding site on a carrier protein

34-	**Saturation of carrier-mediated transport occurs when: (M3WAN - Hard)?**
A-	The carrier protein changes its structure permanently
B-	The substrate binds irreversibly to the carrier
C-	All available carrier proteins are occupied, so increasing substrate concentration no longer increases transport
D-	ATP production stops

35-	**Secondary active transport is defined as the transport of a substance: (M3WAN - Core)?**
A-	Coupled with sodium ion movement through a common carrier protein
B-	Directly coupled to ATP hydrolysis

C-	Independent of any ion gradient
D-	Occurring only through ungated channels

36-	**Which of the following is transported by co-transport (symport) with sodium? (M3WAN - Core)?**		
A-	Hydrogen ion	B-	Calcium ion
C-	None of these are co-transported	D-	Glucose

37-	**Which of the following is transported by counter-transport (antiport) with sodium? (M3WAN - Hard)?**						
A-	Glucose	B-	Amino acids	C-	Hydrogen ion	D-	Chloride

38-	**Secondary active transport and facilitated diffusion are similar in that both: (Important M3WAN - Hard)?**		
A-	Directly use ATP	B-	Use carrier proteins
C-	Transport water only	D-	Use channel proteins exclusively

39-	**Formation of a vesicle from the cell membrane to take up an extracellular particle into the cell is called: (M3WAN - Core)?**		
A-	Endocytosis	B-	Exocytosis
C-	Facilitated diffusion	D-	Osmosis

40-	**Fusion of an intracellular vesicle with the cell membrane and release of its contents to the exterior is called: (M3WAN - Core)?**		
A-	Endocytosis	B-	Simple diffusion
C-	Primary active transport	D-	Exocytosis

41-	**Regarding vesicular transport, which of the following is TRUE? (M3WAN - Hard)?**
A-	It always occurs against the concentration gradient
B-	It has no fixed relation to the concentration gradient and may occur with or against it
C-	It never utilizes ATP
D-	It does not affect the surface area of the cell membrane

42-	**Regarding the effect of vesicular transport on the cell membrane: (M3WAN - Core)?**
A-	Neither endocytosis nor exocytosis affects membrane surface area
B-	Endocytosis adds membrane area while exocytosis removes it
C-	Endocytosis removes areas of membrane while exocytosis adds areas to it
D-	Both endocytosis and exocytosis reduce membrane surface area permanently

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	C	B	A	C	D	B	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	D	C	A	B	D	C	B	A
21	22	23	24	25	26	27	28	29	30
D	C	B	A	D	C	B	A	D	C
31	32	33	34	35	36	37	38	39	40
B	A	D	C	A	D	C	B	A	D
41	42	43	44	45	46	47	48	49	50
B	C	-	-	-	-	-	-	-	-

من اشتغل بطاعة الله تكفل الله برزقه، كما في حديث زيد بن ثابت المرفوع:

»مَنْ كَانَتِ الدُّنْيَا هَمَّهُ فَرَّقَ اللَّهُ عَلَيْهِ أَمْرَهُ، وَجَعَلَ فَقْرَهُ بَيْنَ عَيْنَيْهِ، وَلَمْ يَأْتِهِ مِنَ الدُّنْيَا إِلَّا مَا كُتِبَ لَهُ، وَمَنْ كَانَتِ الْآخِرَةُ نِيَّتَهُ جَمَعَ اللَّهُ لَهُ أَمْرَهُ، وَجَعَلَ غِنَاهُ فِي قَلْبِهِ، وَأَتَتْهُ الدُّنْيَا وَهِيَ رَاغِمَةٌ« خرّجه الإمام أحمد وابن ماجه.

وخرج الترمذي من حديث أنس مرفوعًا:

»إِنَّ اللَّهَ يَقُولُ: يَا ابْنَ آدَمَ، تَفَرَّغْ لِعِبَادَتِي أَمْلَأْ صَدْرَكَ غِنًى، وَأَسُدَّ فَقْرَكَ، وَإِلَّا تَفْعَلْ مَلَأْتُ يَدَيْكَ شُغْلًا، وَلَمْ أَسُدَّ فَقْرَكَ.«

مجموع رسائل ابن رجب ١/٢٤٢

Resting membrane potential

Pre Exams & Formative

1-	**Which isa true about the resting membrane potential in a neuron? (MID 31)**
A-	The membrane potential is closer to the K+ equilibrium potential than to the Na equilibrium potential
B-	The membrane potential is at the equilibrium potential for K+
C-	There is no ion movement during resting membrane potential
D-	There is no ion movement by the Na+/K+ ATPase pump

2-	**A drug toa muscle caused the opening of many Chloride (-) selective channels in the plasma membrane. What change in resting membrane potential will occur in response to this drug? (MID 31)**		
A-	Depolarization	B-	Hyperpolarization
C-	Repolarization	D-	No change

3-	**About ion channel state and correct associated electrical event? (MID 31)**
A-	Open voltage-gated K channels cause the depolarization.
B-	Open voltage-gated K+ channels determines the resting membrane potential
C-	Open K+ leak channel determine RMP
D-	Open voltage-gated Na+ channels determines the resting membrane potential

4-	**Suppose that gated ion channels for Na+ or Ca2+ opened in the plasma membrane of a resting muscle cell. The membrane potential of that cell would? (MID 31)**
A-	Become more negative than the resting membrane potential
B-	Become less positive than the resting membrane potential
C-	Move toward the equilibrium potential for that ion
D-	Move toward the potassium equilibrium potential

5-	**The resting membrane potential of a neuron is? (FINAL 31)**
A-	Equal to the potassium equilibrium potential
B-	Equal to the sodium equilibrium potential
C-	Somewhat less negative than the potassium equilibrium potential
D-	Somewhat more positive than the sodium equilibrium potential

6-	**Which is true about the resting membrane potential in a neuron? (FINAL 31)**
A-	There is passive ions movement during resting membrane potential
B-	The membrane potential is at the equilibrium potential for K+
C-	The membrane potential is at the equilibrium potential for Na+

D-	There is no Na+ /K+ ATPase pump activity

7-	**At rest, passive movement of K+ is more than Na+ due to all of the following EXCEPT? (FINAL 31)**
A-	Conc. gradient for K+ is larger than Conc. gradient for Na+
B-	Number of K+ channels is more than Na+ channels
C-	Size of hydrated K+ ion is less than size of hydrated Na+ ion
D-	Ca2+ ions help K+ ion efflux

8-	**RMP of a nerve? (MID 32, 2ND TRIAL 32, MID 33, FINAL 34)**
A-	is caused by equal distribution of ions along both sides of the membrane
B-	is caused by selective permeability of the membrane to the ions
C-	Na+ -K+ pump has no role in RMP
D-	is caused mainly by inward movement of Na+ ions

9-	**A neuron cell membrane is described as "resting polarized"' when? (FINAL 31)**
A-	It is actively conducting a nerve impulse
B-	The cell is totally incapable of excitation
C-	The intracellular fluid is more negative than extracellular fluid
D-	No ions are crossing through it

10-	**What is largely responsible for the negative resting membrane potential (around -70 mV) ina neuron? (FINAL 32)**		
A-	Axonal insulation by Schwann cells	B-	Voltage-gated sodium channels opening
C-	The action potential	D-	Potassium leak currents

11-	**Which of the following gives the ordering of ions in terms of their ionic concentration inside the typical neuron, from highest to lowest? (FINAL 32)**						
A-	K+, Na+, Ca++	B-	Na+, K+, Ca++	C-	Ca++, Na+, K+	D-	Ca++, K+, Na+

12-	**Resting Membrane Potential (RMP)? (FINAL 36)**		
A-	It is characteristic of excitable tissue	B-	Increase by hypercalcemia
C-	Electrical	D-	None of the above

13-	**The most important factor in maintaining resting membrane potential is: (MID 37)**						
A-	K influx	B-	Na efflux	C-	K efflux	D-	The Na/K pump

14-	**Nernst potential ... (FINAL 37)**
A-	Prove that K ions have the major role in RMP

B-	Equilibrium potential is the Resting membrane potential

15-	**Resting membrane potential of a nerve fiber: (Formative 36)**
A-	Is mainly due to chloride influx
B-	Is mainly due to potassium influx.
C-	is due to the presence of negative charges on the inner surface in relation to its outer surface

16-	**Regarding the resting membrane potential (RMP), what is its main cause? (2ND TRIAL 37)**
A-	Efflux of K^+
B-	Influx of K^+
C-	Influx of Na^+
D-	Efflux of Na^+

17-	**Which of the following is correct about the Na^+-K^+ pump? (2ND TRIAL 37)**
A-	Has a main role in maintaining the RMP
B-	Has no effect on generating the RMP
C-	Pumps 3 K^+ into the cell and 2 Na^+ out of the cell
D-	Does not utilize ATP

Answers - Pre Exams & Formative

1	2	3	4	5	6	7	8	9	10
A	B	C	C	C	A	D	B	C	D
11	12	13	14	15	16	17	18	19	20
A	A&C	D	A	C	A	A	-	-	-

Department Qs

1-	**Resting nerve cell membrane is more permeable to : (SNU Bank, Dr. Mohamed Ashraf)**		
A-	K+ ions than to Na+ ions.	B-	organic anions than to Cl-- ions.
C-	glucose molecules than to oxygen molecules.	D-	calcium ions than to K + ions

2-	**Concerning the nerve fiber membrane at rest, the following is true: (SNU Bank, Dr. Mohamed Ashraf)**
A-	the concentration gradient for K+ is such that it tends to move into fiber.
B-	the Na+-K+ pump moves Na+ in and K+ out of the fiber.
C-	Extracellular K+ concentration is more than its intracellular concentration.
D-	if the resting membrane potential is moved toa less negative value, the fiber becomes more excitable.

3-	Which of the following ions is actively transported out of the neuron? (SNU Bank, Dr. Mohamed Ashraf)						
A-	Na+.	B-	K+.	C-	Cl-.	D-	HCO3.

4-	Na+-K+ pump of neuron: (SNU Bank, Dr. Mohamed Ashraf)
A-	it is the main cause of the resting membrane potential.
B-	it couples extrusion of Na+ out of the cell in exchange with K+ into the cell at a ratio of 2:1.
C-	it requires energy derived from ATP.
D-	activity is increased during descending limb of the spike potential.

5-	Na+-K+ pump: (SNU Bank, Dr. Mohamed Ashraf)
A-	is the main cause for creation of RMP.
B-	is essential for spike formation.
C-	activity increases in proportion to intracellular Na+ concentration.
D-	couples Na+ and K+ pump with a ratio 1:1.

6-	Resting membrane potential in medium sized nerve is normally about: (SNU Bank, Dr. Mohamed Ashraf)								
A-	50 mv.	B-	+ 35 mv.	C-	90 mv.	D-	70 mv.	E-	55 mv.

7-	The main ion responsible for the resting membrane potential is: (SNU Bank, Dr. Mohamed Ashraf)								
A-	Na+.	B-	K+.	C-	Ca++.	D-	Cl-.	E-	HCO3-.

8-	Concerning the RMP: (SNU Bank, Dr. Mohamed Ashraf)
A-	it is mainly due to K+ influx through leak channels.
B-	it is mainly due to Na+ efflux.
C-	it is usually of fixed value in all tissues.
D-	it is mainly due to K+ efflux through leak channels.
E-	it is more negative in nerve fibers than muscle fibers.

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
A	D	A	C	C	D	B	D	-	-

M3WAN Bank

1-	Membrane potential is best defined as: (M3WAN - Core)
A-	The total ionic concentration inside the cell only
B-	The rate of ion diffusion across the membrane

C-	The magnitude of charge difference between the inside and outside of the cell, measured in millivolts
D-	The volume of the intracellular compartment

2-	**The membrane potential of a cell under specific conditions is determined by: (M3WAN - Core)**
A-	The concentration of ions inside/outside the cell and membrane permeability to those ions
B-	Cell size and membrane thickness only
C-	The total body water content
D-	The rate of protein synthesis

3-	**The electrochemical gradient across the neuronal membrane is primarily established by differences in the concentration of: (M3WAN - Core)**		
A-	Glucose, urea, and creatinine	B-	Ca2+, Mg2+, and phosphate only
C-	Oxygen and carbon dioxide	D-	Na+, Cl-, and K+

4-	**Non-diffusible negatively charged intracellular particles that contribute to the electrochemical gradient include: (M3WAN - Hard)**		
A-	Glucose and urea	B-	Proteins and phosphates (as in DNA/ATP)
C-	Sodium and chloride	D-	Oxygen and carbon dioxide

5-	**Regarding the ionic composition of extracellular fluid relative to intracellular fluid: (M3WAN - Core)**
A-	ECF consists mostly of sodium, with small amounts of K+ and Cl-, making the outer surface positive
B-	ECF consists mostly of potassium, making the outer surface negative
C-	ECF and ICF have identical ionic compositions
D-	ECF contains mostly large organic anions

6-	**Regarding the ionic composition of intracellular fluid: (M3WAN - Core)**
A-	It consists mostly of sodium and chloride
B-	It is nearly ion-free
C-	It consists mostly of potassium, with large organic anions such as proteins and phosphates
D-	It contains mostly extracellular-type anions

7-	**The equilibrium (Nernst) potential is defined as the membrane potential at which: (Important M3WAN - Core)**
A-	The membrane is completely depolarized
B-	There is no net diffusion of a particular type of ion
C-	The Na/K pump stops functioning
D-	The cell membrane ruptures

8-	**Passive movement of an ion across the cell membrane is governed by two forces, namely: (M3WAN - Core)**
A-	Osmotic pressure and hydrostatic pressure
B-	Temperature and pH
C-	ATP availability and enzyme activity
D-	The concentration gradient and the electrical gradient

9-	**Regarding the relationship between the concentration gradient of an ion and its equilibrium potential: (M3WAN - Hard)**
A-	The greater the concentration gradient, the greater the equilibrium potential needed to prevent its diffusion
B-	The greater the concentration gradient, the smaller the equilibrium potential
C-	Equilibrium potential is independent of the concentration gradient
D-	Equilibrium potential only depends on the electrical gradient

10-	**If the equilibrium potential for Na+ is +60 mV, this means that: (Important M3WAN - Core)**
A-	The potential inside the cell should be smaller than outside by 60 mV
B-	There is no potential difference across the membrane
C-	The potential inside the cell should be larger than outside by 60 mV
D-	Na+ is in a state of active transport

11-	**If the equilibrium potential for K+ is -90 mV, this means that: (Important M3WAN - Core)**
A-	The potential inside the cell should be larger than outside by 90 mV
B-	The potential inside the cell should be less than outside by 90 mV
C-	There is no potential difference across the membrane
D-	K+ is being actively pumped into the cell

12-	**The resting membrane potential (RMP) is best defined as: (Important M3WAN - Core)**
A-	The peak potential reached during an action potential
B-	The equilibrium potential of sodium
C-	The potential difference recorded only during muscle contraction
D-	The potential difference between the outer and inner surfaces of the membrane during rest

13-	**The normal resting membrane potential value in medium-sized nerves approximates: (Important M3WAN - Core)**						
A-	90 mV	B-	55 mV	C-	70 mV	D-	+35 mV

14-	**The normal resting membrane potential value in large-sized nerves approximates: (M3WAN - Core)**						
A-	90 mV	B-	55 mV	C-	35 mV	D-	70 mV

15-	The resting membrane potential is also known as the: (M3WAN - Core)		
A-	Action potential	B-	Equilibrium potential
C-	Threshold potential	D-	Transmembrane potential

16-	The resting membrane potential is measured usinga: (M3WAN - Core)		
A-	Sphygmomanometer	B-	Cathodal ray oscilloscope
C-	Spectrophotometer	D-	Electroencephalograph

17-	To record a true resting membrane potential using two microelectrodes, they must be placed: (Important M3WAN - Core)
A-	One on the outer surface and the other on the inner surface
B-	Both on the outer surface of the axon
C-	Both on the inner surface of the axon
D-	Both outside the cell in the extracellular fluid

18-	If both recording microelectrodes are placed on the same surface (both outer or both inner) ofa resting axon, the result is: (M3WAN - Core)
A-	A potential difference equal to the RMP
B-	A potential difference equal to the equilibrium potential of Na+
C-	No potential (voltage) difference between the electrodes
D-	An action potential is triggered

19-	Passive factors account for approximately what percentage of the resting membrane potential? (Important M3WAN - Core)						
A-	7%	B-	50%	C-	100%	D-	93%

20-	The Na+/K+ pump (active factor) accounts for approximately what percentage of the resting membrane potential? (Important M3WAN - Core)						
A-	93%	B-	7%	C-	50%	D-	0%

21-	The passive factors responsible for generating the resting membrane potential mainly include: (M3WAN - Core)
A-	Selective permeability with diffusion of Na+ and K+, and non-diffusible intracellular particles
B-	Active transport of glucose and amino acids
C-	Vesicular transport of proteins
D-	Osmotic movement of water only

22-	The resting cell membrane is more permeable to K+ than to Na+ by approximately: (Important M3WAN - Core)						
A-	2 to 5 times	B-	10 to 20 times	C-	50 to 100 times	D-	200 to 500 times

23-	The higher permeability of the resting membrane to K+ compared with Na+ is explained by all of the following EXCEPT: (M3WAN - Hard)
A-	A higher concentration gradient for K+
B-	A greater number of K+ leak channels than Na+ leak channels
C-	Guarding of Na+ channels by calcium ions
D-	Active pumping of K+ by the Na+/K+ ATPase

24-	Sodium leak channels are relatively less permeable partly because they are guarded by: (M3WAN - Hard)		
A-	Potassium ions	B-	Calcium ions, which repel sodium
C-	Chloride ions	D-	Magnesium ions

25-	Compared with potassium, sodium is less permeable through leak channels partly because: (M3WAN - Hard)
A-	The hydrated sodium ion is thicker than the hydrated potassium ion
B-	Sodium has no charge
C-	Sodium has a smaller hydration shell than potassium
D-	Sodium leak channels are more numerous than potassium leak channels

26-	The hydration energy of an ion is: (M3WAN - Hard)		
A-	Directly proportional to its atomic number	B-	Unrelated to its atomic number
C-	Inversely proportional to its atomic number	D-	Equal for all monovalent ions

27-	Regarding the role of the Na+/K+ pump in the resting membrane potential: (Important M3WAN - Core)
A-	It plays a great role in generation and a small role in maintenance
B-	It plays no role in either generation or maintenance
C-	It plays an equal role in generation and maintenance
D-	It plays a small role in generation and a great role in maintenance

28-	The Na+/K+ pump contributes to the resting membrane potential mainly because it is: (M3WAN - Core)
A-	A non-electrogenic pump that has no effect on membrane charge
B-	An electrogenic pump that pumps 3 Na+ out and 2K+ in, generating net outward positive current
C-	A channel protein that allows free diffusion of both ions
D-	Only active during an action potential

29-	**Regarding the historical use of the equilibrium potential to explain the resting membrane potential: (M3WAN - Hard)**
A-	It failed, since the measured RMP differs from the equilibrium potential of any single ion type
B-	It succeeded completely, as RMP exactly equals the equilibrium potential of Na+
C-	It succeeded only for calcium
D-	It was never attempted historically

30-	**Comparing the measured RMP with the equilibrium potentials of individual ions shows that the most important ion in RMP generation is: (Important M3WAN - Core)**
A-	Sodium, since its equilibrium potential is nearest to RMP
B-	Chloride, since it is most abundant in ECF
C-	Potassium, since its equilibrium potential is nearest to RMP
D-	Calcium, since it guards sodium channels

Answers - M3WAN Bank

1	2	3	4	5	6	7	8	9	10
C	A	D	B	A	C	B	D	A	C
11	12	13	14	15	16	17	18	19	20
B	D	C	A	D	B	A	C	D	B
21	22	23	24	25	26	27	28	29	30
A	C	D	B	A	C	D	B	A	C

من العقوبات التي رتّبها الله على الذنوب :

١ –الختمُ على القلوب والأسماع، والغشاوةُ على الأبصار، والإقفالُ على القلوب .

٢ –التثبيط عن الطاعة، والإقعادُ عنها .

٣ –جعلُ القلبِ أصمَّ لا يسمعُ الحق، أبكم لا ينطقُ به، أعمى لا يراه .

٤ –الخسفُ بالقلب كما يُخسفُ بالمكان وما فيه، فيُخسفُ به إلى أسفلِ السافلين، وصاحبهُ لا يشعر .

٥ –البُعد عن البرِّ والخيرِ ومعالي الأعمال .

٦ –مسخُ القلب .

٧ –مكرُ الله بالماكر، ومخادعتُهُ للمخادع واستهزاؤهُ بالمستهزئ، وإزاغتُهُ لقلب الزائغ عن الحق .

•ابن القيم | الداء والدواء صـ(١٧٠)

Action potential

Pre Exams & Formative

1-	**Which of these isa character of action potentials? (MID 31)**
A-	Na+ andK+ gates begin to open at the same time
B-	It has same shape, duration in all cells
C-	They are produced by ligand gated channels
D-	The membrane potential reverses polarity during repolarization
E-	None of the above

2-	**About ion channel state and associated electrical event? (FINAL 31)**
A-	Open voltage-gatedK+ channels cause after depolarization
B-	Voltage-gated Na+ channels are inactivated ina resting neuronal membrane
C-	Open voltage-gatedK+ channels cause the depolarization
D-	Open voltage-gatedK+ channels determine the resting membrane potential

3-	**At threshold of Action Potential ,which is correct? (FINAL 31)**
A-	Inner gate ofK+ Voltage gated channel is opened
B-	Inner gate in Na+ Voltage gated channel is open
C-	Outer Gate in Na+ Voltage gated channel is closed
D-	Inner gate in Na+ Voltage gated channel is closed

4-	**Which of these isa characteristic of action potentials? (FINAL 31)**
A-	They occur in all cells
B-	They are produced by Ligand gated channels
C-	The membrane potential reverses polarity during depolarization
D-	Na+ andK+

5-	**Repolarization? (2ND TRIAL 32, FINAL 34)**
A-	Occurs at first gradual then becomes fast
B-	Results from closure of sodium gates and opening of potassium gates
C-	is represented by the ascending limb of the spike
D-	is followed by appearance of response

6-	**depolarization of the nerves occurs by opening of? (2ND TRIAL 32)**		
A-	voltage gated sodium channels	B-	Mechanically gatedK channels
C-	Ligand gated Cl channels	D-	Leak channels

7-	**Depolarization is? (MID 33)**
A-	associated with increase in membrane permeability to Na+
B-	terminated with closure of voltage activatedK + channels
C-	followed by muscle relaxation
D-	caused byK+ efflux

8-	**Action potential of nerve fiber? (2ND TRIAL 35)**
A-	Results from transient reversal of polarity of membrane potential
B-	Terminated by influx ofK
C-	Initiated by efflux of Na
D-	Decline in magnitude as it moves along nerve fiber

9-	**After hyperpolarization stage is due to? (FINAL 36)**		
A-	Slow closure ofK+ gates	B-	Ligand gated channels
C-	Leak channels	D-	None of the above

10-	**The most important channel during action potential? (FINAL 36)**						
A-	Electrical	B-	Ligand gated	C-	Leak	D-	Voltage gated

11-	**During positive after potential phase of action potential, the membrane is: (MID 37)**		
A-	More excitable	B-	near to firing level
C-	Hypopolarized	D-	less excitable

12-	**At the start of repolarization of an action potential, voltage gated Na channels are: (MID 37)**		
A-	Opened	B-	rapidly closed by inner gate
C-	closed by outer gate	D-	slowly closing by inner gate

13-	**The all or none law refers to the: (MID 37)**
A-	Amount of released Na andk during nerve spike potential
B-	Amplitude of the nerve spike potential
C-	Magnitude of resting membrane potential
D-	Duration of nerve spike potential

14-	**Repolirazation spirke potential (FINAL SNU 1)**		
A-	Na diffusing Rapidly	B-	K+ Rapidly outside
C-	K+Rapidly inside	D-	K+ Slowly outside

15-	What happened during the descending line of action potential? (FINAL SNU 1)		
A-	K+ outflux through voltage-gatedK+ channels	B-	K+ outflux through leak channels
C-	K+ influx through voltage-gatedK+ channels	D-	K+ influx through leak channels

16-	The response for maintain the concentration of Na after Active Potential? (FINAL 37)		
A-	Na/Kpump	B-	Voltage gated Na channel

17-	The action potential of nerve:- (Formative 36)
A-	is not associated with any movement of Na+ orK+ across the membrane
B-	Declines in amplitude as it moves along the axon.
C-	Is terminated byK+ efflux.
D-	Is initiated by Na+ efflux.

18-	During the upstroke of the nerve action potential (Formative 37)
A-	there is net inward current and the cell interior becomes more negative
B-	there is net outward current and the cell interior becomes more negative
C-	there is net inward current and the cell interior becomes less negative
D-	there is net outward current and the cell interior becomes less negative

19-	The all-or-none law applies to: (2ND TRIAL 37)
A-	Duration of the latent period
B-	Amplitude of the resting membrane potential
C-	Speed of conduction
D-	Amplitude of the action potential

20-	At –55 mV during the depolarization phase, which of the following occurs? (2ND TRIAL 37)
A-	K^+ voltage-gated channels open slowly
B-	Na^+ voltage-gated channels open slowly
C-	K^+ voltage-gated channels open rapidly
D-	Na^+ voltage-gated channels open rapidly

21-	During the rapid repolarization phase of the spike potential: (2ND TRIAL 37)
A-	Na^+-gated channels open rapidly
B-	K^+-gated channels open slowly
C-	K^+-gated channels are closed
D-	K^+-gated channels open rapidly

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
E	A	B	C	B	A	A	A	A	D
11	12	13	14	15	16	17	18	19	20
D	B	B	B	A	A	C	C	D	D
21	22	23	24	25	26	27	28	29	30
D	-	-	-	-	-	-	-	-	-

Department Qs

1-	During the depolarization phase of the nerve action potential: (SNU Bank, Dr. Mohamed Ashraf)
A-	there is an increased influx of bothK+ and Cl- ions.
B-	the membrane becomes more permeable toK+ than Na+ ions.
C-	there is an increased influx of Na+ ions.
D-	Na+ channels are blocked by Ca++ ions.

2-	Regarding the local response ina nerve fibre, the following is true, EXCEPT: (SNU Bank, Dr. Mohamed Ashraf)
A-	it can result from application of subthreshold stimulus.
B-	it is proportional to the strength of the stimulus.
C-	it can propagate along the nerve fiber for long distance.
D-	it is accompanied by increased excitability.

3-	The membrane potential will depolarize by the greatest amount if the membrane permeability increases for: (SNU Bank, Dr. Mohamed Ashraf)		
A-	potassium.	B-	sodium.
C-	sodium and potassium.	D-	chloride.

4-	During absolute refractory period : (SNU Bank, Dr. Mohamed Ashraf)
A-	K+ channels are inactivated.
B-	Na+ channels are inactivated by inner gates.
C-	stronger stimulus higher than threshold can excite the nerve fiber.
D-	coincides with whole duration of the action potential.

5-	The action potential: (SNU Bank, Dr. Mohamed Ashraf)
A-	can be increased in size if the stimulus strength is increased.
B-	is smaller in size if the extracellular Na+ concentration is reduced.

C-	hasa depolarization phase which corresponds to increasedK+ influx.
D-	is of shorter duration in skeletal muscle than in nerve fiber.

6-	**Concerning the nerve fibre: (SNU Bank, Dr. Mohamed Ashraf)**
A-	at rest, the inside of the nerve is positive to the outside.
B-	the resting potential becomes more negative if the Na+ permeability of the membrane rises.
C-	during the rising phase of the action potential, there is an increase in efflux ofK+ ions.
D-	the descending limb of the action potential is largely due to increasedK+ efflux.

7-	**The action potential of a neuron: (SNU Bank, Dr. Mohamed Ashraf)**
A-	is initiated by efflux of Na+.
B-	is terminated by efflux of K+.
C-	decreases in amplitude as it moves along the axon.
D-	results in a transient reversal of concentration gradient of Na+ across the cell membrane

8-	**Voltage-gated Na+ channels: (SNU Bank, Dr. Mohamed Ashraf)**
A-	have activation gates on the inner surface of the membrane and inactivation gates on the outer surface.
B-	inactivation gates are opened at rest.
C-	activation gates open during catelectrotonus.
D-	at threshold potential, the inactivation gates close.

9-	**Action potential: (SNU Bank, Dr. Mohamed Ashraf)**
A-	has magnitude dependent on the stimulus intensity.
B-	is accompanied by increasing nerve excitability.
C-	velocity is decreased with myelination.
D-	is caused by transient increase in Na+ permeability followed by a prolonged increase in K+ permeability.

10-	**During an action potential, depolarization of nerve cell membrane is brought about by: (SNU Bank, Dr. Mohamed Ashraf)**		
A-	increased sodium permeability.	B-	increased calcium permeability.
C-	increased chloride permeability.	D-	increased potassium permeability.

11-	**Depolarization in the nerve: (SNU Bank, Dr. Mohamed Ashraf)**
A-	is caused by Na+ outflow.
B-	is caused by K+ outflow.
C-	is due to opening of inactivation gates of voltage gated Na+ channels.
D-	is represented by descending limb of the spike.

E-	is that during which the nerve can not respond to any stimulus.

12-	Descending limb of the spike potential in the nerve: (SNU Bank, Dr. Mohamed Ashraf)		
A-	represents depolarization.	B-	is caused by potassium inflow.
C-	is caused by sodium inflow.	D-	represents Repolarization.
E-	is caused by calcium inflow.		

13-	During depolarization of nerve fiber: (SNU Bank, Dr. Mohamed Ashraf)
A-	sodium ions move to the outside due to activation of voltage gated sodium channels.
B-	excitability decreases markedly.
C-	potassium ions move to the outside due to activation of voltage gated potassium channels.
D-	the membrane potential changes from - 90 mv to maximally - 10 mv.
E-	bothc &d are correct.

14-	The ascending limb of the spike potential in the nerve: (SNU Bank, Dr. Mohamed Ashraf)		
A-	is caused byK+ inflow.	B-	represents depolarization.
C-	is caused by Na+ inflow through leak channels.	D-	represents Repolarization.
E-	is associated with increased excitability.		

15-	Threshold for production of an action potential ina nerve: (SNU Bank, Dr. Mohamed Ashraf)
A-	isa membrane potential (transmembrane voltage) at which the increase inK+ permeability exceeds the increase in Na+ permeability.
B-	isa condition where Na+ channels are triggered to open normally.
C-	isa condition where active Na+ transport stops.
D-	a&b are correct.
E-	none of the above.

16-	The stimulation of myelinated nerve fibers would produce: (SNU Bank, Dr. Mohamed Ashraf)		
A-	outward flow of sodium ions.	B-	inward flow of potassium ions.
C-	reversal of membrane potential.	D-	displacement of myelin sheath.
E-	none of the above are correct.		

17-	Local response: (Dr. Mohamed Ashraf)
A-	isa state of passive depolarization at the anode.
B-	it is accompanied by increased excitability.
C-	it obeys all or non law.
D-	it can propagate and generate another local response within a distance of 3-4 cm.

18-	The local response: (Dr. Mohamed Ashraf)		
A-	is characterized by depressed excitability.	B-	cannot be summated.
C-	obeys the all or none law.	D-	is followed by an absolute refractory period.
E-	none of the above.		

19-	Which of the following is correct: (Dr. Mohamed Ashraf)
A-	the amplitude of Action Potential depends on stimulus strength.
B-	graded (local) potentials can only cause membrane depolarization.
C-	the amplitude of action potential does not depend on stimulus strength.
D-	saltatory conduction occurs in unmyelinated nerve fibers.
E-	conduction of Action Potential in the axon occurs only in one direction.

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	C	B	B	B	D	B	B	D	A
11	12	13	14	15	16	17	18	19	20
E	D	B	B	B	C	B	E	C	-

M3WAN Bank

1-	An action potential is best defined as: (M3WAN - Core)
A-	The steady-state potential difference across a resting membrane
B-	A rapid, brief, transient change in membrane potential following stimulation of an excitable cell by an adequate stimulus
C-	A permanent change in resting membrane potential
D-	A slow, prolonged hyperpolarization of the membrane

2-	The voltage-gated Na+ channel possesses: (M3WAN - Core)
A-	Asingle activation gate only
B-	Two inactivation gates only
C-	An outer fast activation gate and an inner slow inactivation gate
D-	No gates, beinga leak channel

3-	Passage of Na+ through the voltage-gated Na+ channel requires: (M3WAN - Core)
A-	Both the activation and inactivation gates to be open
B-	Only the activation gate to be open
C-	Only the inactivation gate to be open

D-	Neither gate needs to be open

4-	**The "closed but capable of opening" state of the voltage-gated Na+ channel is characterized by: (M3WAN - Hard)**		
A-	Activation gate open, inactivation gate closed	B-	Both gates open
C-	Both gates closed and incapable of opening	D-	Activation gate closed, inactivation gate open

5-	**The "open (activated)" state of the voltage-gated Na+ channel is characterized by: (M3WAN - Core)**
A-	Activation gate closed, inactivation gate open
B-	Activation gate open, inactivation gate closed
C-	Both the activation and inactivation gates open
D-	Both gates closed

6-	**The "inactivated" state of the voltage-gated Na+ channel is characterized by: (M3WAN - Hard)**
A-	Activation gate closed, inactivation gate open, capable of opening
B-	Activation gate open, inactivation gate closed, and the channel is not capable of opening
C-	Both gates open, actively conducting Na+
D-	Both gates closed, capable of opening immediately

7-	**Compared with the voltage-gated Na+ channel, the voltage-gated K+ channel has: (M3WAN - Core)**
A-	Two gates, similar to the Na+ channel
B-	Three gates
C-	No gates at all
D-	Only one gate (activation gate), which can be closed or open

8-	**The stimulus artifact recorded on application of a stimulus is caused by: (M3WAN - Hard)**
A-	Leakage of current from the stimulating electrode to the detecting electrode in the fluid outside the axon
B-	Opening of voltage-gated Na+ channels
C-	Closure of K+ channels
D-	Activation of the Na+/K+ pump

9-	**The significance of the stimulus artifact is that it indicates: (M3WAN - Core)**
A-	The amplitude of the resulting action potential
B-	The time of application of the stimulus
C-	The conduction velocity of the axon
D-	The refractory period duration

10-	**The latent period is defined as the time between: (M3WAN - Core)**
A-	The peak of depolarization and the peak of repolarization
B-	Two successive action potentials
C-	The after depolarization and after hyperpolarization
D-	Application of the stimulus (stimulus artifact) and the beginning of the action potential curve

11-	**The latent period is used to measure: (M3WAN - Core)**		
A-	The resting membrane potential value	B-	The amplitude of the action potential
C-	The conduction speed of the axon	D-	The number of voltage-gated channels

12-	**Conduction speed of an axon can be calculated as: (M3WAN - Hard)**
A-	Distance between stimulating and recording electrodes divided by the latent period
B-	Latent period divided by distance between electrodes
C-	Distance multiplied by the latent period
D-	Amplitude of the action potential divided by distance

13-	**During slow depolarization from -70 mV to -63 mV, the initial change is mainly due to: (M3WAN - Hard)**
A-	Massive opening of voltage-gated Na+ channels
B-	Complete inactivation of Na+ channels
C-	Closure of K+ leak channels
D-	Addition of negative charge to the outer surface of the membrane from the stimulating electrode

14-	**During slow depolarization from -63 mV to -55 mV, the change is mainly due to: (M3WAN - Core)**
A-	Complete closure of all Na+ channels
B-	Opening of a few voltage-gated Na+ channels, which further depolarizes the membrane and opens more channels
C-	Maximal K+ efflux
D-	Activation of the Na+/K+ pump

15-	**The positive feedback cycle in which opening of a few voltage-gated Na+ channels leads to further depolarization and opening of more channels is called the: (M3WAN - Core)**						
A-	Hodgkin cycle	B-	Nernst cycle	C-	Starling cycle	D-	Krebs cycle

16-	**The threshold for stimulation (firing level) of a typical nerve fiber approximates: (Important M3WAN - Core)**						
A-	70 mV	B-	90 mV	C-	55 mV	D-	+35 mV

17-	Rapid and irreversible depolarization at the threshold level is mainly due to: (Important M3WAN - Core)
A-	Rapid opening of all voltage-gated K+ channels
B-	Rapid opening of all voltage-gated Na+ channels, causing massive Na+ influx
C-	Complete closure of Na+ leak channels
D-	Activation of the Na+/K+ pump

18-	At the point of complete depolarization, the potential difference between the outer and inner surfaces of the membrane is: (M3WAN - Core)		
A-	70 mV	B-	+35 mV
C-	90 mV	D-	Zero (isopotential)

19-	The overshoot phase of the action potential refers to: (Important M3WAN - Core)
A-	Reversal of polarity from zero to about +35 mV, with the inside becoming positive relative to the outside
B-	A return of the membrane potential to -70 Mv
C-	A drop of the membrane potential to -90 mV
D-	A plateau at -55 mV

20-	The peak of the action potential approximates: (M3WAN - Core)						
A-	55 mV	B-	70 mV	C-	+35 mV	D-	90 mV

21-	At the peak of the action potential (+35 mV): (Important M3WAN - Core)
A-	Na+ channels open maximally and Na+ continues to enter rapidly
B-	K+ channels close completely
C-	The Na+/K+ pump reverses its direction
D-	Na+ channels become inactivated and no further Na+ can enter the cell

22-	Rapid repolarization represents approximately what portion of the total repolarization process? (M3WAN - Core)		
A-	The first 30%	B-	The first 70%
C-	The last 30%	D-	100% of the process

23-	The combination of rapid depolarization and rapid repolarization is together called the: (M3WAN - Core)						
A-	After potential	B-	Local response	C-	Spike potential	D-	Stimulus artifact

24-	Rapid repolarization is mainly caused by: (Important M3WAN - Core)
A-	Closure of Na+ channels together with increased K+ outflow
B-	Continued Na+ influx together with K+ channel closure

C-	Activation of the Na+/K+ pump alone
D-	Complete closure of all ion channels

25-	**Maximal opening of voltage-gated K+ channels occurs at approximately the same time as: (M3WAN - Hard)**		
A-	Opening of Na+ channels at -55 mV	B-	Inactivation of Na+ channels at +35 mV
C-	The resting state at -70 mV	D-	The after hyperpolarization at -90 mV

26-	**The after-potential phase of the action potential occurs: (M3WAN - Core)**
A-	Before depolarization begins
B-	Simultaneously with the overshoot
C-	During the latent period only
D-	After the first 70% of the repolarization process, when the membrane potential changes slowly

27-	**During the negative after potential (after-depolarization), the membrane is: (Important M3WAN - Core)**
A-	Hypopolarized and more excitable, being closer to the firing level
B-	Hyperpolarized and less excitable
C-	At the exact resting potential with normal excitability
D-	Completely in excitable

28-	**The negative after-potential (after-depolarization) is mainly caused by: (M3WAN - Hard)**
A-	Increased outward diffusion of K+
B-	Complete closure of Na+ channels
C-	Decreased outward diffusion of K+ due to loss of some of the electrical driving force
D-	Massive activation of the Na+/K+ pump

29-	**During the positive after-potential (after-hyperpolarization), the membrane potential approximates: (Important M3WAN - Core)**
A-	55 mV, and the membrane is hypopolarized and more excitable
B-	0 mV, and the membrane is isopotential
C-	+35 mV, and the membrane is at its peak
D-	90 mV, and the membrane is hyperpolarized and less excitable

30-	**The positive after-potential (after-hyperpolarization) is mainly caused by: (M3WAN - Hard)**
A-	Rapid closure of Na+ channels
B-	Slow closure of K+ channels with continued K+ efflux from the cell
C-	Continued Na+ influx into the cell

D-	Sudden activation of ligand-gated channels

31-	**After an action potential, the role of the Na+/K+ pump is to: (M3WAN - Core)**
A-	Pump the small amount of Na+ that entered back out, and the small amount of K+ that left back in
B-	Pump Na+ into the cell and K+ out of the cell
C-	Stop all ionic movement permanently
D-	Generate the overshoot phase

32-	**The all or none law states that the action potential: (Important M3WAN - Core)**
A-	Increases in amplitude as stimulus strength increases above threshold
B-	Occurs only with subthreshold stimuli
C-	Occurs with a constant amplitude and shape irrespective of the magnitude of a supra-threshold stimulus
D-	Varies in shape depending on stimulus duration

33-	**Regarding suprathreshold stimuli and the all or none law: (M3WAN - Core)**
A-	Suprathreshold stimuli produce larger action potentials
B-	Suprathreshold stimuli produce longer-duration action potentials
C-	Suprathreshold stimuli fail to produce any action potential
D-	Action potentials produced by suprathreshold stimuli have the same amplitude, duration, and form as those produced by threshold stimuli

34-	**The mechanism underlying the all or none law is that: (Important M3WAN - Core)**
A-	Voltage-gated K+ channels determine the amplitude of the action potential
B-	Maximum activation of voltage-gated Na+ channels occurs at the firing level, beyond which additional stimulus intensity has no further effect on Na+ influx
C-	The Na+/K+ pump determines the amplitude of the action potential
D-	Ligand-gated channels determine the threshold level

Answers - M3WAN Bank

1	2	3	4	5	6	7	8	9	10
B	C	A	D	C	B	D	A	B	D
11	12	13	14	15	16	17	18	19	20
C	A	D	B	A	C	B	D	A	C
21	22	23	24	25	26	27	28	29	30
D	B	C	A	B	D	A	C	D	B
31	32	33	34	35	36	37	38	39	40
A	C	D	B	-	-	-	-	-	-

Physiological Properties of Nerve Fibers

Pre Exams & Formative

1-	**As the strength of a stimulus to an axon is increased? (MID 31)**
A-	The amplitude of action potentials increases
B-	The action potential remains the same
C-	The duration of action potentials increases
D-	The speed with which action potentials are conducted increases

2-	**Ifa nerve cell is said to be more excitable, this mean? (FINAL 31)**
A-	It needs a strong stimulus
B-	Its membrane potential is closer to its threshold
C-	It will produce a large action potential
D-	It has a longer action potential

3-	**Chronaxie (FINAL 34)**
A-	is the minimal stimulus which produce response after relatively long period
B-	is the minimal time needed by the strong stimulus to produce response
C-	is the time required to stimulate the nerve by a minimal stimulus
D-	is the time required to stimulate the nerve by a stimulus which is double rheobase.

4-	**One of the following is true about absolute refractory period? (FINAL 34)**
A-	It follows relative refractory period
B-	It is due to closure of potassium channels
C-	It is the period during which the nerve can be stimulated
D-	It is due to inactivation of sodium channels

5-	**In all or non rule? (FINAL 35)**
A-	Minimal stimulus produces minimal response
B-	A minimal stimulus produces a maximal response
C-	The nerve trunk either respond maximally or not respond at all
D-	The response in a single nerve fiber increases with increase intensity of stimulus

6-	**In all or none rule? (2ND TRIAL 35)**
A-	Minimal stimulus produces minimal response
B-	A minimal stimulus produces a maximal response
C-	The nerve trunk either responds maximally or not at all

D-	The response ina single nerve fiber increases with increase intensity of stimulus

7-	**Chronaxie is? (2ND TRIAL 35)**
A-	The time needed for the rheobase to stimulate a nerve fiber
B-	The minimum duration of a stimulus needed to excite the nerve
C-	A measurement of excitability
D-	The utilization time

8-	**Excitability of neuron is increased by? (2ND TRIAL 35)**						
A-	Hypocalcemia	B-	Hyponatremia	C-	Hypokalemia	D-	Hypomagnesemia

9-	**The time needed to excite a nerve by a current strength equal to the double rheobase is called: (MID 37)**						
A-	utilization time	B-	minimal time	C-	chronaxie	D-	minimal strength

10-	**A decrease in extracellular Calcium (Ca2+) concentration leads to: (FINAL SNU 1)**
A-	Hyperpolarization of the nerve membrane.
B-	An increase in the membrane permeability to Sodium (Na+).
C-	Inhibition of neurotransmitter release.
D-	Decreased nerve excitability.

11-	**The chronaxie:- (Formative 36)**
A-	is twice the rheobase
B-	is the threshold stimulus
C-	is the time needed to excite a nerve by a current strength equal to the rheobase.
D-	Can be used as a measure of excitability.

12-	**Regarding the strength-duration curve, which is true? (2ND TRIAL 37)**
A-	A- Chronaxie is inversely proportional to excitability
B	B- Rheobase is the minimum time required to produce an impulse
C-	C- Chronaxie is the minimum strength of a stimulus required to excite the nerve
D-	D- Rheobase is the time needed to stimulate the nerve with double the threshold strength

Answers - Pre Exams & Formative									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**B**	**D**	**D**	**B**	**B**	**A**	**A**	**C**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	**A**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

Department Qs

1-	for excitation of the nerve fiber, we need at least: (SNU Bank, Dr. Mohamed Ashraf)		
A-	subthreshold stimulus.	B-	threshold stimulus.
C-	suprathreshold stimulus.	D-	maximal stimulus.

2-	Excitability of the nerve fiber is decreased in the following condition: (SNU Bank, Dr. Mohamed Ashraf)		
A-	A decrease in ECF Na+ concentration.	B-	A decrease in ECF K+ concentration.
C-	an increase in ECF K+ concentration.	D-	low Ca++ concentration in ECF.

3-	Excitability of the nerve fiber is increased in the following condition: (SNU Bank, Dr. Mohamed Ashraf)		
A-	low Ca++ concentration in ECF.	B-	high Ca++ concentration in ECF.
C-	A decrease of Na+ concentration in ECF.	D-	A decrease in extracellular K+ concentration.

4-	Hypokalemia reduces excitability of the nerve and muscle cells because decreased extracellular K+: (SNU Bank, Dr. Mohamed Ashraf)
A-	increases activity of Na+-K+ ATPase which hyperpolarizes the cell.
B-	hyperpolarizes the cell, which increases the action potential threshold.
C-	decreases the number of Na+ channels.
D-	depolarizes the cell, thus inactivating voltage-gated Na+ channels.

5-	Regarding excitability of a nerve: (SNU Bank, Dr. Mohamed Ashraf)		
A-	is maximum at the firing level.	B-	is high during repolarization.
C-	can respond toa second stimulus during ARP.	D-	is zero during RRP.

6-	The Strength-Duration curve: (SNU Bank, Dr. Mohamed Ashraf)
A-	is the relationship between the duration of the stimulus and amplitude of response.
B-	Has a rheobase which is 2 times chronaxie.
C-	Has a chronaxie which is the time needed by double the rheobasic strength to stimulate the nerve.
D-	strong stimuli of extreme short duration can excite the nerve.

7-	Chronaxie is: (SNU Bank, Dr. Mohamed Ashraf)
A-	the time needed for the rheobase to stimulate a nerve fiber.
B-	the minimum duration of a stimulus needed to excite the nerve.
C-	A measurement of excitability.
D-	the utilization time.

8-	Excitability of the nerve is increased by: (SNU Bank, Dr. Mohamed Ashraf)				
A-	hypercalcemia.	B-	hypokalemia.	C-	acidosis.
D-	hypocalcemia.	E-	cooling.		

9-	The absolute refractory period: (SNU Bank, Dr. Mohamed Ashraf)
A-	isa period during which excitability is completely lost.
B-	isa period during which stronger stimulus is needed to excite the muscle fiber.
C-	is coincided with the descending limb of the spike.
D-	isa period during which the excitability is less than normal.
E-	none of the above.

10-	It is correct to say that: (SNU Bank, Dr. Mohamed Ashraf)
A-	subthreshold stimulus leads to an action potential.
B-	A decrease in Ca++ in ECF leads toa decrease in excitability in the nerve.
C-	A decrease in K+ in ECF leads to an increase in excitability in the nerve.
D-	absolute refractory period is associated with loss of excitability.
E-	absolute refractory period is associated with only a decrease in excitability.

11-	Relative refractory period: (SNU Bank, Dr. Mohamed Ashraf)
A-	is an increase in excitability.
B-	isa complete loss of excitability.
C-	coincides with the second 2/3 of descending limb of spike.
D-	during it the nerve can't respond to any stimulus whatever its strength.
E-	coincides with after hyperpolarization.

12-	Nerve excitability is increased by: (SNU Bank, Dr. Mohamed Ashraf)				
A-	cooling.	B-	local anaesthetics	C-	acidosis.
D-	deep pressure.	E-	alkalosis.		

13-	All or none law can be applied to: (SNU Bank, Dr. Mohamed Ashraf)		
A-	whole skeletal muscle.	B-	nerve trunk.
C-	single nerve fiber.	D-	multiunit smooth muscle.

14-	The excitability of the nerve is zero during: (SNU Bank, Dr. Mohamed Ashraf)		
A-	super normal phase of excitability.	B-	relative refractory period.
C-	sub normal phase of excitability.	D-	local response.
E-	absolute refractory period.		

15-	The term "chronaxie" refers to: (SNU Bank, Dr. Mohamed Ashraf)
A-	the utilization time.
B-	the minimal duration of a stimulus which can excites the nerve.
C-	the time needed by the rheobase to stimulate the nerve fiber.
D-	the time needed by the current of twice the rheobase intensity to stimulate the nerve.
E-	none of the above.

16-	Chronaxie is: (SNU Bank, Dr. Mohamed Ashraf)		
A-	the duration of the rheobase stimulus.	B-	double rheobase stimulus.
C-	the utilization time.	D-	the rheobase stimulus.
E-	inversely proportional to excitability.		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	A&B	A	B	A	C	A	D	A	D
11	12	13	14	15	16	17	18	19	20
C	E	C	E	D	E	-	-	-	-

M3WAN Bank

1-	Excitability of nerve fibers refers to their ability to: (M3WAN - Core)
A-	Respond to stimuli and generate electrical impulses
B-	Propagate impulses to muscles and glands only
C-	Resist all forms of stimulation
D-	Store electrical charge indefinitely

2-	Conductivity of nerve fibers refers to: (M3WAN - Core)
A-	Generation of the impulse at the site of stimulation only
B-	The fiber's resistance to electrical current
C-	Propagation of the generated electrical impulse along the entire length of the fiber and to other neurons, muscles, or glands
D-	The fiber's ability to metabolize glucose

3-	The refractory period of a nerve fiber refers to the fact that: (M3WAN - Core)
A-	The fiber can conduct multiple simultaneous action potentials
B-	The fiber can conduct only one action potential at a time, with reduced excitability during conduction
C-	The fiber becomes permanently inexcitable after one impulse
D-	The fiber's excitability increases during conduction

4-	**Regarding the all or none response as a property of nerve fibers: (M3WAN - Core)**
A-	Increasing stimulus strength increases the size of the action potential
B-	Subthreshold stimuli always generate an action potential
C-	The response varies according to stimulus duration only
D-	A threshold stimulus generates a full action potential, and increasing stimulus strength further does not change it

5-	**Summation in nerve fibers refers to the phenomenon in which: (Important M3WAN - Core)**
A-	Multiple sub-threshold stimuli applied in rapid succession generate an action potential
B-	Asingle sub-threshold stimulus generates an action potential
C-	Asingle supra-threshold stimulus fails to generate an action potential
D-	Repeated stimuli always cancel each other out

6-	**Regarding fatigue in nerve fibers: (Important M3WAN - Core)**
A-	Nerve fibers fatigue rapidly with continuous stimulation
B-	Nerve fibers fatigue only in the presence of anesthetics
C-	Nerve fibers do not become fatigued even when stimulated continuously
D-	Nerve fibers fatigue only during summation

7-	**Regarding the effect of temperature on nerve excitability: (Important M3WAN - Core)**
A-	Cooling decreases nerve excitability
B-	Cooling increases nerve excitability
C-	Temperature has no effect on nerve excitability
D-	Cooling abolishes the refractory period

8-	**Regarding the effect of temperature on nerve excitability: (M3WAN - Core)**		
A-	Warming decreases nerve excitability	B-	Warming increases nerve excitability
C-	Warming has no effect on nerve excitability	D-	Warming abolishes conductivity

9-	**Mechanical pressure applied on a nerve: (M3WAN - Core)**		
A-	Increases its excitability	B-	Has no effect on excitability
C-	Permanently abolishes conductivity	D-	Reduces its excitability

10-	**Nerve excitability is decreased in cases of: (M3WAN - Core)**		
A-	Ischemia	B-	Alkalosis
C-	Hypercalcemia induced hyperexcitability	D-	Increased blood supply

11-	**Lack of oxygen supply toa nerve fiber: (M3WAN - Core)**
A-	Increases nerve excitability
B-	Has no effect on excitability
C-	Decreases nerve excitability
D-	Abolishes the resting membrane potential immediately

12-	**Regarding the effect of H+ concentration on nerve excitability: (Important M3WAN - Core)**		
A-	Alkalinity decreases nerve excitability	B-	Alkalinity increases nerve excitability
C-	Acidity increases nerve excitability	D-	H+ concentration has no effect on excitability

13-	**Regarding the effect of H+ concentration on nerve excitability: (M3WAN - Core)**		
A-	Acidity increases nerve excitability	B-	Alkalinity decreases nerve excitability
C-	H+ concentration has no effect on excitability	D-	Acidity decreases nerve excitability

14-	**Catelectrotonus (the region near the cathode of an applied current): (M3WAN - Hard)**		
A-	Increases nerve excitability	B-	Decreases nerve excitability
C-	Has no effect on excitability	D-	Abolishes the refractory period

15-	**Anelectrotonus (the region near the anode of an applied current): (M3WAN - Hard)**		
A-	Increases nerve excitability	B-	Has no effect on excitability
C-	Decreases nerve excitability	D-	Abolishes conductivity

16-	**Nerve excitability is decreased by all of the following EXCEPT: (M3WAN - Core)**		
A-	Excess CO2	B-	Alcohol
C-	Anesthetic drugs such as ether and chloroform	D-	Alkalinity

17-	**Decreased extracellular Ca2+ concentration: (Important M3WAN - Core)**
A-	Increases membrane permeability to Na+ and decreases the depolarization needed to trigger an action potential, thus increasing excitability
B-	Decreases membrane permeability to Na+ and increases excitability
C-	Has no effect on Na+ permeability
D-	Directly hyperpolarizes the membrane

18-	**Increased extracellular Ca2+ concentration: (Important M3WAN - Core)**
A-	Increases membrane permeability to Na+ and increases excitability
B-	Has no effect on Na+ permeability
C-	Decreases membrane permeability to Na+ and increases the depolarization needed to trigger an action potential, thus decreasing excitability

D-	Directly depolarizes the membrane

19-	**Increased extracellular Na+ concentration: (M3WAN - Core)**
A-	Delays depolarization and decreases excitability
B-	Facilitates the process of depolarization and increases nerve excitability
C-	Has no effect on excitability
D-	Causes immediate hyperpolarization

20-	**Decreased extracellular Na+ concentration: (M3WAN - Core)**
A-	Facilitates depolarization and increases excitability
B-	Has no effect on excitability
C-	Causes immediate action potential generation
D-	Delays the process of depolarization and decreases nerve excitability

21-	**Increased extracellular K+ concentration: (M3WAN - Core)**
A-	Favors K+ influx, leading to depolarization and increased nerve excitability
B-	Favors K+ efflux, leading to hyperpolarization and decreased excitability
C-	Has no effect on excitability
D-	Directly blocks Na+ channels

22-	**Decreased extracellular K+ concentration: (Important M3WAN - Core)**
A-	Favors K+ influx, leading to depolarization and increased excitability
B-	Has no effect on excitability
C-	Favors K+ efflux, leading to hyperpolarization and decreased nerve excitability
D-	Directly blocks Ca2+ channels

23-	**Among the factors determining the effectiveness ofa stimulus, subthreshold stimuli: (M3WAN - Core)**
A-	Always initiate a full action potential
B-	Produce only local responses and fail to initiate an action potential
C-	Produce action potentials of reduced amplitude
D-	Produce action potentials of prolonged duration

24-	**If the intensity of a stimulus is increased very slowly (gradually), the nerve fails to respond because of the property of: (Important M3WAN - Hard)**						
A-	Summation	B-	Refractoriness	C-	Facilitation	D-	Accommodation

25-	**Regarding the duration of a stimulus and its effectiveness, the strength-duration curve shows: (M3WAN - Core)**
A-	A reciprocal relationship between current strength and the duration of flow required to produce an impulse
B-	A directly proportional relationship between current strength and duration required
C-	No relationship between current strength and duration
D-	A fixed duration regardless of current strength

26-	**According to the strength-duration curve, within limits: (M3WAN - Core)**
A-	The stronger the current, the longer the duration required
B-	Strength and duration are always equal
C-	The stronger the current, the shorter the duration required to produce an impulse
D-	Duration is independent of current strength

27-	**According to the strength-duration curve: (M3WAN - Hard)**
A-	Any duration is sufficient for excitation if the strength is high enough
B-	There isa minimal duration needed for excitation, below which no excitation occurs whatever the stimulus strength
C-	There is no minimal duration required
D-	Excitation always occurs regardless of duration

28-	**Rheobase is defined as: (Important M3WAN - Core)**
A-	The duration of current flow required for excitation at twice the threshold strength
B-	The maximal strength of current a nerve can tolerate
C-	The time between two successive action potentials
D-	The minimal strength (threshold intensity) of a galvanic current that can set up an impulse

29-	**The time required for excitation when using the rheobase strength is called the: (M3WAN - Hard)**						
A-	Utilization time	B-	Chronaxie	C-	Latent period	D-	Refractory time

30-	**Chronaxie is defined as: (Important M3WAN - Core)**
A-	The minimal strength of current that can set up an impulse
B-	The time required for excitation at rheobase strength
C-	The duration of current flow required for excitation when using a strength equal to twice the rheobase
D-	The total duration of the action potential

31-	**Regarding the use of chronaxie as an index of excitability: (Important M3WAN - Core)**
A-	The shorter the chronaxie, the lower the excitability of the tissue

B-	The shorter the chronaxie, the greater the excitability of the tissue
C-	Chronaxie has no relation to tissue excitability
D-	The longer the chronaxie, the greater the excitability

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
A	C	B	D	A	C	A	B	D	A
11	12	13	14	15	16	17	18	19	20
C	B	D	A	C	D	A	C	B	D
21	22	23	24	25	26	27	28	29	30
A	C	B	D	A	C	B	D	A	C
31	32	33	34	35	36	37	38	39	40
B	-	-	-	-	-	-	-	-	-

من أقوال السلف في الإخلاص :

قال الجنيد :الإخلاص سرٌّ بين الله وبين العبد ، لا يعلمه مَلَك فيكتبه، ولا شيطان فيفسده، ولا هوًى فيميله .

وقيل لسهلٍ :أيُّ شيءٍ أشدُّ على النفس فقال: الإخلاص، لأنه ليس لها فيه نصيبٌ .

وقال بعضهم :الإخلاص أن لا تطلب على عملك شاهدًا غير الله، ولا مجازيًا سواه .

وقال مكحولٌ :ما أخلص عبدٌ قطُّ أربعين يومًا إلا ظهرت ينابيع الحكمة من قلبه على لسانه .

وقال يوسف بن الحسين :أعزُّ شيءٍ في الدُّنيا: الإخلاص، وكم أجتهد في إسقاط الرِّياء عن قلبي فكأنَّه ينبت على لونٍ آخر .

وقال أبو سليمان الدارانيُّ :إذا أخلص العبد انقطع عنه كثرة الوساوس والرِّياء .

•ابن القيم | مدارج السالكين (٢/٣٥٠)

Type of AP and conduction of the nerve

Pre Exams & Formative

1-	**Action potential? (MID 32)**
A-	isa graded potential
B-	is produced by sub threshold stimulus.
C-	starts with repolarization caused by outward movement of Cl-.
D-	is conducted slower in thin nerve fibers

2-	**As regard conduction of action potential ina nerve? (FINAL 32, 2ND TRIAL 32, FINAL 34)**
A-	in thick myelinated nerve fibers can reach up to 120m/second
B-	can be increased by increase calcium
C-	can be increased by cooling
D-	is conducted with decrement

3-	**All describe slow Conduction except (FINAL 34)**				
A-	Demylinated	B-	Unmylinated	C-	Thick

4-	**Saltatory Conduction (FINAL 34)**
A-	Occurs in unmylinated nerve fibers
B-	may reach up to 120meter/sec
C-	occurs by jumping from one neuron to another
D-	decreases gradually with distance till it disappears.

5-	**Local circuit mechanism for conduction of an action potential: (MID 37)**		
A-	Occurs in type B nerves.	B-	called saltatory conduction
C-	Very rapid	D-	occurs in type C nerve fibers

6-	**To record a Monophasic Action Potential in a nerve axon, the electrodes should be placed: (FINAL SNU 1)**
A-	Both on the external surface of the nerve axon.
B-	Both within the nerve trunk.
C-	One electrode ICF and one electrode ECF nerve axon.
D-	One electrode ICF and one electrode ECF nerve axon nerve trunks.

7-	**Regarding myelinated nerve fibers all are true, except:- (Formative 36)**
A-	Action potential occurs at the node of Ranvier.

B-	They are large fibers
C-	Faster than unmyelinated due to saltatory conduction.
D-	Require more energy than unmyelinated fibers.

8-	**In an action potential, conduction velocity increases with? (2ND TRIAL 37)**
A-	Decreased axon diameter
B-	Increased thickness of the myelin sheath
C-	Absence of nodes of Ranvier
D-	Continuous conduction

9-	**Which of the following is correct about a monophasic action potential recording? (2ND TRIAL 37)**
A-	One recording electrode is placed outside and the other inside the nerve axon
B-	One recording electrode is placed outside and the other inside the nerve trunk
C-	Two electrodes are placed inside the nerve
D-	Two electrodes are placed outside the nerve

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
D	A	C	B	D	C	D	B	A	-

Department Qs

1-	**Regarding conduction of action potential along a nerve fiber: (SNU Bank, Dr. Mohamed Ashraf)**
A-	propagation of action potential requires ATP.
B-	large myelinated nerve fibers conduct faster than unmyelinated nerve fibers.
C-	saltatory conduction is associated with slowing of action potential propagation.
D-	the action potential magnitude declines as it propagates along the nerve fiber.

2-	**All of the following are true for myelin sheath EXCEPT: (SNU Bank, Dr. Mohamed Ashraf)**		
A-	it provides insulation.	B-	it gives protection.
C-	it isa lipoprotein material.	D-	it transmits nerve impulses.
E-	it is present in the myelinated nerve fibers.		

3-	**It is correct to say that: (SNU Bank, Dr. Mohamed Ashraf)** **معلومة من مادة الهستولوجي**
A-	dendrites conduct impulses away from the soma.

B-	axon conducts impulse toward the soma.
C-	neurilemmal sheath is important for nerve regeneration.
D-	myelin is formed of single layer of Schwann cells.

4-	**Saltatory conduction of action potential: (SNU Bank, Dr. Mohamed Ashraf)**
A-	occurs in the unmyelinated nerves.
B-	is slow.
C-	occurs in the sarcolemma.
D-	needs more energy than point to point conduction.
E-	is fast.

5-	**Local circuit mechanism for conduction of action potential: (SNU Bank, Dr. Mohamed Ashraf)**
A-	is more rapid than saltatory conduction.
B-	occurs in the myelinated nerve fibers.
C-	needs less energy than saltatory conduction.
D-	needs more energy than saltatory conduction.
E-	occurs in both myelinated and unmyelinated nerve fibers.

6-	**In saltatory conduction, the nerve impulse: (SNU Bank, Dr. Mohamed Ashraf)**
A-	passes from one node to another node.
B-	shows continuous conduction.
C-	velocity is slower than that of the unmyelinated nerve fiber.
D-	none of the above.

7-	**Propagation of action potential in myelinated nerve fiber is: (SNU Bank, Dr. Mohamed Ashraf)**		
A-	slow.	B-	point to point.
C-	saltatory.	D-	passive.
E-	similar to that of unmyelinated fibers.		

8-	**Which of the following is correct for nerve fibers: (SNU Bank, Dr. Mohamed Ashraf)** معلومة من العملي
A-	C fibers are myelinated.
B-	B fibers are faster than A fibers.
C-	postganglionic nerve fibers are of B type.
D-	preganglionic nerve fibers are of C type.
E-	deep pressure decreases conduction mainly in A type.

9-	**which of the following is the outermost layer that covers the axon of the nerve cell: (Dr. Mohamed Ashraf)** **معلومة من مادة الهستولوجي**		
A-	cell membrane.	B-	myelin sheath.
C-	neurilemmal sheath.	D-	neurofibrils.

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	D	C	E	D	A	C	E	C	-

M3WAN Bank

1-	**To record a monophasic action potential, the two recording electrodes should be placed: (M3WAN - Core)**
A-	Both in the extracellular fluid
B-	One in the extracellular fluid and the other in the intracellular fluid
C-	Both in the intracellular fluid
D-	Both outside the nerve fiber entirely

2-	**For biphasic action potential recording, the two electrodes are placed: (M3WAN - Core)**
A-	Both in the extracellular fluid, or both in the intracellular fluid
B-	One in the extracellular fluid and the other in the intracellular fluid
C-	Both must be inside the myelin sheath
D-	One proximal and one at the neuromuscular junction only

3-	**In biphasic recording from two points (A and B) on the outer surface of a nerve fiber, before the stimulus reaches either point: (M3WAN - Core)**
A-	Point A is already negative relative to point B
B-	Point B is already negative relative to point A
C-	There is no potential difference because both points are isoelectric (positively charged)
D-	Both points show maximal depolarization

4-	**In biphasic recording, when the impulse reaches point A first: (M3WAN - Core)**
A-	Point A becomes hyperpolarized, producing a downward deflection
B-	No deflection is recorded until the impulse reaches point B
C-	Both points depolarize simultaneously
D-	Point A becomes depolarized and negative relative to point B, producing an upward deflection

5-	**In biphasic recording, when the impulse leaves point A: (M3WAN - Core)**
A-	Point A remains permanently depolarized
B-	Point A repolarizes, becoming positive again, and the deflection returns to the baseline
C-	A downward deflection is recorded
D-	The recording stops entirely

6-	**In biphasic recording, when the impulse reaches point B: (Important M3WAN - Core)**
A-	Point B produces an upward deflection identical to point A
B-	No further deflection is recorded
C-	Point B becomes hyperpolarized
D-	Point B becomes depolarized and negative relative to point A, producing a deflection in the opposite (downward) direction

7-	**In biphasic recording, when the impulse leaves point B: (M3WAN - Core)**
A-	Point B repolarizes, becoming positive again, and the two points become isoelectric with the deflection returning to baseline
B-	Point B remains negative permanently
C-	A second upward deflection is recorded
D-	The baseline is never regained

8-	**In unmyelinated nerve fibers, nerve impulses are propagated in the form of: (M3WAN - Core)**
A-	A single instantaneous jump from the origin to the terminal
B-	Saltatory conduction from node to node
C-	A wave of action potentials, with reversal of polarity at each point of stimulation
D-	Direct chemical diffusion of neurotransmitter along the axon

9-	**The flow of current between an activated point and neighboring inactive areas of the nerve membrane is called: (M3WAN - Core)**		
A-	Local circular current	B-	Saltatory current
C-	Equilibrium current	D-	Electrogenic current

10-	**The initial area of negativity produced by an action potential, into which positive charges flow from neighboring inactive areas, is called the: (M3WAN - Hard)**						
A-	Firing level	B-	Current sink	C-	Node of Ranvier	D-	Isoelectric point

11-	**The gradual decrease in membrane polarity at inactive areas, produced by local circular currents flowing from an active area, is called: (Important M3WAN - Core)**		
A-	Saltatory depolarization	B-	Overshoot

C-	After-hyperpolarization	D-	Electrotonic depolarization

12-	**In unmyelinated fibers, once an inactive area is electrotonically depolarized to the firing level: (M3WAN - Core)**
A-	It generates its own action potential and, in turn, electrotonically depolarizes the next area, propagating the impulse to the end of the fiber
B-	The impulse stops, since unmyelinated fibers cannot propagate impulses
C-	The impulse jumps directly to the nerve terminal
D-	The impulse travels back toward the point of origin only

13-	**The special mechanism of impulse propagation along myelinated nerve fibers, in which depolarization jumps from node to node, is called: (M3WAN - Core)**		
A-	Local circuit conduction	B-	Continuous conduction
C-	Saltatory conduction	D-	Electrotonic conduction

14-	**Regarding the myelin sheath surrounding the nerve axon: (M3WAN - Core)**
A-	It acts as an insulator to current flow
B-	It has high permeability to Na+ and easily permits current flow
C-	It generates action potentials directly
D-	It is the main site of impulse generation

15-	**Compared with the myelin segments, the nodes of Ranvier: (M3WAN - Core)**
A-	Act as insulators to current flow
B-	Have no ion channels at all
C-	Are impermeable to all ions
D-	Have high permeability to Na+ and easily permit current flow

16-	**In saltatory conduction, positive charges jump from: (M3WAN - Core)**
A-	One myelin segment directly to the next myelin segment
B-	The inactive nodes to the area of current sink at the active node, bypassing the insulating myelin segments
C-	The cell body to the axon terminal instantaneously
D-	The postsynaptic membrane to the presynaptic membrane

17-	**One of the advantages of saltatory conduction is: (M3WAN - Core)**
A-	An increase in the velocity of conduction
B-	A decrease in the velocity of conduction
C-	Complete abolition of the refractory period
D-	Elimination of the need fora threshold stimulus

18-	Saltatory conduction conserves energy mainly because: (Important M3WAN - Core)
A-	No ATP is required anywhere along a myelinated fiber
B-	The Na+/K+ pump is inactive in myelinated fibers
C-	Excitation (and thus active ion pumping) occurs only at the nodes and not all over the nerve membrane
D-	Myelin itself generates ATP for the axon

19-	The conduction velocity in very small unmyelinated nerve fibers can be as low as: (M3WAN - Core)						
A-	2.5m/sec	B-	25m/sec	C-	0.025m/sec	D-	0.25m/sec

20-	The conduction velocity in very large myelinated nerve fibers can reach as high as: (Important M3WAN - Core)						
A-	10m/sec	B-	100m/sec	C-	1m/sec	D-	1000m/sec

21-	Erlanger and Gasser classified nerve fibers into groups: A,B, and C mainly on the basis of: (Important M3WAN - Core)
A-	The type of neurotransmitter released
B-	The presence or absence of a myelin sheath only, without reference to diameter or velocity
C-	The mutual relationship between nerve fiber diameter and conduction velocity
D-	The anatomical location of the fiber in the CNS versus PNS

22-	According to the Erlanger-Gasser classification, nerve fibers are grouped into: (M3WAN - Core)		
A-	Three major groups: A, B, and C	B-	Two groups: myelinated and unmyelinated only
C-	Four groups: A, B, C, and D	D-	Five groups based on function

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	A	C	D	B	D	A	C	A	B
11	12	13	14	15	16	17	18	19	20
D	A	C	A	D	B	A	C	D	B
21	22	23	24	25	26	27	28	29	30
C	A	-	-	-	-	-	-	-	-

ANS I (Introduction & Ganglia)

Pre Exams & Formative

1-	The parasympathetic neurons are originated from the following parts of the spinal cord? (MID 31, 2ND TRIAL 32, Formative 36)		
A-	Sacral spinal cord	B-	Thoracic spinal cord
C-	Cervical spinal cord	D-	Lumbar spinal cord

2-	The parasympathetic division of the autonomic nervous system is characterized by? (FINAL 31, MID 32)		
A-	Adrenergic postganglionic fibers	B-	Its origin is only from spinal cord
C-	Short postganglionic fibers	D-	None of the above is correct

3-	The sympathetic system originates from? (FINAL 31)						
A-	T1 to L3	B-	T2 to L1	C-	C1 to L2	D-	T1 to S2

4-	parasympathetic ganglia are located? (MID 33, FINAL 34, FINAL 35)		
A-	Ina chain parallel to the spinal cord	B-	Dorsal root ganglia
C-	Next to or within the organs innervate	D-	Craniosacral

5-	The neurotransmitter of preganglionic sympathetic fibers is? (2ND TRIAL 35)						
A-	Norepinephrine	B-	Epinephrine	C-	Acetylcholine	D-	Dopamine

6-	The preganglionic sympathetic nerve that innervate the eye originate from: (FINAL 37)		
A-	Superior colliculi	B-	Thoracic segments of spinal cord
C-	Superior cervical ganglion	D-	Ciliary ganglion

7-	Which of the following is correct about the parasympathetic nervous system? (2ND TRIAL 37)
A-	Has a craniosacral outflow
B-	Arises from the lateral horn cells (LHCs) of the thoracolumbar segments
C-	Supplies the adrenal medulla
D-	Its postganglionic fibers are long

Answers - Pre Exams & Formative									
1	2	3	4	5	6	7	8	9	10
A	C	A	C	C	B	A	-	-	-

Department Qs

1-	**The major difference between the autonomic & somatic divisions of the nervous system is : (SNU Bank, Department Qs, Dr.Seham)**
A-	Only the somatic nerves innervate skeletal muscle.
B-	The somatic nerves only originate from the thoraco-lumbar segments of the spinal cord.
C-	The autonomic nerves only originate from the thoraco-lumbar segments of the spinal cord.
D-	Only the autonomic nerves innervate blood vessels.

2-	**The autonomic N'S. differs from the somatic N'S. in having: (SNU Bank, Department Qs, Dr.Seham)**
A-	A connector neuron which comes outside the CNS (preganglionic neuron) &a postganglionic neuron which supplies the organ.
B-	Origin from all segments of the spinal cord.
C-	A double innervation to every organ.
D-	A complete control on skeletal muscles.

3-	**The autonomic ganglia: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Are 5 types.	B-	Are the site of relay of afferent neurons.
C-	Function as distributing centers.	D-	Are located inside the CNS.

4-	**Preganglionic autonomic fibres are of: (SNU Bank, Department Qs, Dr.Seham)**						
A-	Type: A(alpha)	B-	Type: A(delta)	C-	Type: B	D-	Type: C

5-	**The sympathetic division of the autonomic nervous system is characterized by: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Localized actions affecting few systems.	B-	Thoracolumbar outflow from the spinal cord.
C-	Short postganglionic fibres	D-	Adrenergic preganglionic fibres

6-	**The symp. preganglionic nerve fibres: (SNU Bank, Department Qs, Dr.Seham)**
A-	Leave the spinal cord in the grey rami communicantes.
B-	Originate at the autonomic ganglia.
C-	Are myelinated nerve fibres belonging to the B group of nerve fibres.
D-	Show minimal divergence in the autonomic ganglia.

7-	**The following reflexes are all autonomic EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**		
A-	Salivary secretion.	B-	Micturition.
C-	Flexion of the arm following a painful stimulus.	D-	Defecation.

8-	**The autonomic nervous system differs from the somatic nervous system in: (SNU Bank, Department Qs, Dr.Seham)**
A-	Having one efferent neuron.
B-	Preganglionic neurons are located in the anterior horn of the spinal cord.
C-	Efferent nerves have either excitatory or inhibitory effects on the effector organs.
D-	Postganglionic neurons are located in the lateral horn of spinal cord.

9-	**The autonomic ganglia have all the following characters EXCEPT: (SNU Bank, Department Qs, Dr.Seham)**
A-	Are three types: lateral, collateral and terminal.
B-	Are sites of relay of autonomic preganglionic and postganglionic fibres.
C-	Are all present adjacent to the spinal cord.
D-	Are functioning as distributing centres.

10-	**The parasympathetic division is characterized by? (SNU Bank)**		
A-	Adrenergic postganglionic fibers	B-	Its origin is only from spinal cord
C-	Short postganglionic fibers	D-	None of the above

11-	**Sympathetic system originates from? (SNU Bank)**						
A-	T1 to L3	B-	T2 to L1	C-	C1 to L2	D-	T1 to S2

12-	**The parasympathetic neurons are originated from the following parts of the spinal cord? (SNU Bank)**				
A-	Sacral spinal cord	B-	Cervical spinal cord	C-	Thoracic spinal cord
D-	Lumbar spinal cord	E-	All of the above are wrong		

13-	**The neurotransmitter of preganglionic sympathetic fibers is? (SNU Bank)**						
A-	Norepinephrine	B-	Epinephrine	C-	Acetylcholine	D-	Dopamine

14-	**Stimulation of the sympathetic N'S' causes: (Department Qs, Dr.Seham)**
A-	Contraction of the ciliary muscle for near vision
B-	Generalized actions affecting many systems
C-	Bronchoconstriction
D-	Decreased glycogenolysis & lipolysis

15-	**Concerning the autonomic ganglia, all the following is true EXCEPT: (Department Qs, Dr.Seham)**
A-	They acting as distributing centers
B-	They are 3 types
C-	Their chemical transmitter is acetylcholine

D-	Their receptors are specifically blocked by atropine

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
A&D	A	C	C	B	C	C	C	C	C
11	12	13	14	15	16	17	18	19	20
A	A	C	B	D	-	-	-	-	-

M3WAN Bank

1-	Body activities are controlled mainly by two systems, namely: (M3WAN - Core)		
A-	The endocrine system and the immune system	B-	The nervous system and the skeletal system
C-	The endocrine system and the nervous system	D-	The digestive system and the nervous system

2-	The endocrine system controls body activity mainly through: (M3WAN - Core)		
A-	Slow action by hormones	B-	Rapid action by neurotransmitters
C-	Instantaneous mechanical signals	D-	Direct electrical coupling between organs

3-	The nervous system controls body activity mainly through: (M3WAN - Core)		
A-	Slow action by hormones	B-	Slow diffusion of metabolites
C-	Passive osmotic signals	D-	Rapid action by neurotransmitters

4-	The central nervous system (CNS) is composed of: (M3WAN - Core)		
A-	The brain and the cranial nerves only	B-	The brain and the spinal cord
C-	The spinal cord and the autonomic ganglia	D-	The cerebral cortex and the peripheral nerves

5-	The spinal cord is divided into a total of: (M3WAN - Core)						
A-	24 segments	B-	12 segments	C-	31 segments	D-	42 segments

6-	Regarding the segmental composition of the spinal cord: (M3WAN - Hard)
A-	8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal
B-	12 cervical, 8 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal
C-	7 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 2 coccygeal
D-	8 cervical, 5 thoracic, 12 lumbar, 5 sacral, and 1 coccygeal

7-	The peripheral nervous system-scranial nerves consist of a total of: (M3WAN - Core)						
A-	10 pairs	B-	31 pairs	C-	8 pairs	D-	12 pairs

8-	The dorsal root of a spinal nerve carries: (M3WAN - Core)		
A-	Motor (efferent) fibers	B-	Sensory (afferent) fibers
C-	Both sensory and motor fibers equally	D-	Only autonomic preganglionic fibers

9-	The ventral root of a spinal nerve carries: (M3WAN - Core)		
A-	Motor (efferent) fibers	B-	Sensory (afferent) fibers
C-	Only postganglionic autonomic fibers	D-	Purely parasympathetic fibers

10-	The somatic nervous system is characterized by being: (M3WAN - Core)
A-	Involuntary
B-	Purely sensory, with no motor component
C-	Voluntary
D-	Controlled entirely by the spinal cord independent of the brain

11-	The autonomic nervous system is characterized by being: (M3WAN - Core)		
A-	Voluntary	B-	Involuntary
C-	Controlled directly by skeletal muscle	D-	Purely sensory

12-	The somatic motor system consists of: (Important M3WAN - Core)
A-	A chain of two motor neurons with a ganglion in between
B-	Three successive motor neurons
C-	No motor neurons, only sensory relay
D-	One motor neuron extending from the CNS directly to skeletal muscle

13-	Axons of the somatic motor system are: (M3WAN - Core)
A-	Well myelinated and conduct impulses rapidly
B-	Thinly myelinated or unmyelinated, conducting slowly
C-	Devoid of any myelin, conducting extremely rapidly
D-	Present only within autonomic ganglia

14-	The autonomic motor pathway consists of: (Important M3WAN - Core)
A-	Asingle motor neuron extending directly to the effector organ
B-	Three motor neurons in series
C-	A chain of two motor neurons: a preganglionic neuron and a postganglionic neuron
D-	A motor neuron with no synapse outside the CNS

15-	Compared with the somatic motor system, conduction in the autonomic nervous system is: (M3WAN - Core)
A-	Faster, due to heavily myelinated axons
B-	Slower, due to thinly myelinated or unmyelinated axons

C-	Identical in speed to the somatic system
D-	Instantaneous, since no synapse is involved

16-	**The autonomic nervous system is also known as the: (M3WAN - Core)**		
A-	Central or voluntary nervous system	B-	Peripheral sensory nervous system
C-	Somatosensory nervous system	D-	Visceral or involuntary nervous system

17-	**The autonomic nervous system contributes to homeostasis through regulation of all of the following EXCEPT: (M3WAN - Core)**		
A-	Voluntary contraction of skeletal muscle	B-	Blood pressure
C-	Gastrointestinal responses to food	D-	Contraction of the urinary bladder

18-	**The sympathetic nervous system arises from: (Important M3WAN - Core)**
A-	The lateral horn cells of the 2nd, 3rd, and 4th sacral segments (craniosacral outflow)
B-	Nuclei of cranial nerves III, VII, IX, and X only
C-	The lateral horn cells of all thoracic and the upper 3 lumbar segments of the spinal cord (thoracolumbar outflow)
D-	The dorsal horn cells of all spinal segments

19-	**The parasympathetic nervous system arises from: (Important M3WAN - Core)**
A-	The lateral horn cells of all thoracic and upper 3 lumbar segments (thoracolumbar outflow)
B-	The dorsal root ganglia of all spinal segments
C-	The cerebral cortex exclusively
D-	The lateral horn cells of the 2nd, 3rd, and 4th sacral segments, together with nuclei of cranial nerves III, VII, IX, and X (craniosacral outflow)

20-	**The autonomic pathway is formed of: (M3WAN - Core)**
A-	Only preganglionic fibers, with no ganglia
B-	Preganglionic nerve fibers, autonomic ganglia, and postganglionic nerve fibers
C-	Asingle continuous neuron with no relay
D-	Postganglionic fibers only, arising directly from the spinal cord

21-	**An autonomic ganglion is defined as: (M3WAN - Core)**
A-	A collection of nerve cells outside the central nervous system
B-	A collection of nerve cells inside the central nervous system
C-	A single specialized neuron within the spinal cord
D-	A bundle of purely sensory fibers

22-	A collection of nerve cells located inside the central nervous system is known as a: (M3WAN - Core)						
A-	Ganglion	B-	Plexus	C-	Center or nucleus	D-	Ramus

23-	The chemical transmitter present in all autonomic ganglia is: (Important M3WAN - Core)						
A-	Noradrenaline	B-	Dopamine	C-	Serotonin	D-	Acetylcholine

24-	Autonomic ganglia function mainly as sites of: (M3WAN - Core)
A-	Direct hormone secretion into the bloodstream
B-	Relay (synapse) between preganglionic and postganglionic fibers
C-	Storage of red blood cells
D-	Generation of the resting membrane potential

25-	Lateral (paravertebral) ganglia are: (Important M3WAN - Core)
A-	Purely sympathetic
B-	Purely parasympathetic
C-	Relay both sympathetic and parasympathetic fibers
D-	Not part of the autonomic nervous system

26-	The lateral (paravertebral) ganglia forming the sympathetic chain number a total of: (Important M3WAN - Core)						
A-	31 ganglia	B-	12 ganglia	C-	23 ganglia	D-	45 ganglia

27-	Regarding the cervical lateral ganglia, unlike other spinal segments which have one pair of ganglia each: (Important M3WAN - Hard)
A-	The cervical ganglia are absent entirely
B-	The cervical region has one pair per segment, like all other regions
C-	The cervical ganglia are purely parasympathetic
D-	The cervical ganglia are collected into only 3 pairs

28-	Collateral ganglia are located: (M3WAN - Core)
A-	At both sides of the vertebral column
B-	Between the sympathetic chain and the viscera, at the origin of major arteries from the abdominal aorta
C-	Inside the wall of the organ they supply
D-	Exclusively within the brainstem

29-	**Examples of collateral ganglia include the celiac, renal, and superior/inferior mesenteric ganglia, named according to: (M3WAN - Core)**
A-	The blood vessels at which they are located
B-	The spinal segment from which their preganglionic fibers arise
C-	The neurotransmitter they release
D-	The organ they directly penetrate

30-	**Unlike lateral and terminal ganglia, collateral ganglia are unique in that they: (Important M3WAN - Core)**
A-	Relay only sympathetic preganglionic fibers
B-	Relay only parasympathetic preganglionic fibers
C-	Relay both sympathetic and parasympathetic preganglionic fibers
D-	Do not relay any preganglionic fibers

31-	**Terminal ganglia are located: (Important M3WAN - Core)**
A-	At both sides of the vertebral column
B-	At the origin of major arteries from the abdominal aorta
C-	Exclusively within the spinal cord
D-	Near the organ of supply or inside its wall

32-	**Terminal ganglia are: (Important M3WAN - Core)**
A-	Purely sympathetic, receiving fibers to the skin and blood vessels
B-	Purely parasympathetic, receiving preganglionic fibers to the bladder, rectum, or sex organs
C-	Mixed sympathetic and parasympathetic relay stations
D-	Not associated with any pelvic organs

33-	**One of the functions of autonomic ganglia as distributing centers is that: (Important M3WAN - Core)**
A-	Each preganglionic sympathetic fiber gives rise to about 20 postganglionic fibers, leading toa widespread effect
B-	Each preganglionic fiber synapses with only one postganglionic fiber
C-	Ganglia have no distributing function
D-	Postganglionic fibers always outnumber preganglionic fibers by a factor of 2

34-	**Autonomic ganglia are clinically significant as the sites of action of: (M3WAN - Core)**		
A-	Beta-blocker drugs acting on the heart only	B-	Drugs that act exclusively on skeletal muscle
C-	Ganglion-blocking drugs	D-	Drugs that block only postganglionic receptors

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	A	D	B	C	A	D	B	A	C
11	12	13	14	15	16	17	18	19	20
B	D	A	C	B	D	A	C	D	B
21	22	23	24	25	26	27	28	29	30
A	C	D	B	A	C	D	B	A	C
31	32	33	34	35	36	37	38	39	40
D	B	A	C	-	-	-	-	-	-

تم بفضل الله ، هذا وما كان من توفيق فمن الله وحده وما كان من خطأ أو زلل فمن أنفسنا ومن الشيطان ، وفقنا الله وإياكم لخيري الدنيا والآخرة وأعاننا على طاعته .

نسألكم الدعاء لكل من سبقنا من إخواننا إلى الله أن يتغمدهم بواسع رحمته وأن يرفع قدرهم ويتجاوز عن سيئاتهم وأن يجمعنا بهم في جناته .